

## Research Article

# Potential of a Nonperennial Tributary Integrated with Solar Energy for Rural Electrification: A Case Study of Ikukwa Village in Tanzania

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This study evaluates the hydropower potential in the design of a micro-hydro/solar photovoltaic hybrid system with battery energy storage for increasing the access to electricity in Ikukwa Village in Mbeya Region of Tanzania. Usually, hybridized hydropower schemes are designed from perennial streams for the provision of electricity. This study incorporates the run-of-the river (COE) power scheme, which originates from the untapped potential of nonperennial hydro-energy source and the use of traditional approach of data measurements for Ikata tributary to design hybrid system. The system is optimized by the minimization of the total net present cost (NPC) and cost of energy (COE) using the soft computing method of Hybrid Optimization of Multiple Energy Resources (HOMER) software and artificial intelligent (AI) techniques. AI optimization techniques such as particle swarm optimization (PSO), grey wolf optimization (GWO), and GWO-PSO hybrid (GWO-PSOHD) algorithms have been employed for further optimal results. The data for solar radiation and the tributary have been obtained from the National Aeronautics and Space Administration (NASA) and traditional methods of measurements, respectively. The estimated maximum water flow rate and head are 2.943 m<sup>3</sup>/s and 13 m, respectively. In the same period, the approximated theoretical power potential of the tributary is found to be 375 kW. Total NPCs obtained from HOMER, PSO, GWO, and GWO-PSOHD methods are \$ 141,397.76, \$ 95,167.21, \$ 92,472.82, and \$ 91,854.10, respectively. Similarly, the optimal results of COE from HOMER, PSO, GWO, and GWO-PSOHD approaches are \$ 0.1818/kWh, \$ 0.1185/kWh, \$ 0.1182/kWh, and \$ 0.1181/kWh, respectively. Comparatively, PSO implementation has indicated the greatest energy cost, while the cost acquired by GWO-PSOHD is the lowest for all aforementioned AI optimization techniques. The tributary under study has a high potential of diversification of energy sources for rural electrification in the area of study and other parts of the world with comparable conditions.

## 1. Background

Energy is one of the greatest significant infrastructures needed for prosperous human life. Electricity improves the standard of living and economic growth of any country. Electricity supports the development of agriculture, industry, transport, tourism, and mining [1]. The majority of countries globally construct fossil fuel-driven power plants to meet energy demand through on- and off-grid energy systems. Sources of electricity are hydro, natural gas, and liquid fuels [2]. For the past at least twenty years, an

estimated 80% of the Tanzanian power sector has been dominated by hydro-energy source [3, 4]. The total capacity of hydropower plants (HHPs) was equal to 561.84 MW. Capacities for off-grid NGPP and off-grid diesel power plants (DPPs) before were 25.25 MW and 28.682 MW, respectively. However, the addition of natural gas power plants (NGPPs) with the estimated capacity of around 495.44 MW surpassed the magnitude of hydropower source [3]. The approximated total capacity of electricity is 1450 MW of which NG, HPP, and NGPP correspond to 652.5 MW, 609 MW, and 188.5 MW, respectively [5]. Fossil fuels used in

electric power generation for grid extension are depletive and expensive. These fossil fuels are also detrimental to the environment causing pollution, global warming, and other numerous abnormalities [6]. The long gestation period of implementing the national electric grid and its power quality issues associated with the power delivery from distant generating places delays to avail electricity in off-grid locations [7]. Electrification in rural areas is lower than that in urban locations [3]. The decentralized approach via diesel-powered generators has been used to generate electricity in order to minimize the power deficit in the rural areas [5, 8]. However, diesel generators are also not preferable because of the high fuel price and carbon oxide emissions, which are harmful to the environment [8]. Therefore, it is advisable to include renewable energy sources in the power generation sector of Tanzania [5]. Renewable energy sources found in rural areas can be utilized to meet energy demand [9] since the country has a great potential of renewable energy resources, which can solve energy deficiency in the rural areas. These energy resources comprise hydro, wind, biomass, solar, geothermal, and tidal and wave [10, 11]. These renewable energy sources could have helped much to address power catastrophe in the nation, particularly in the sectors of agriculture, mining, tourism, industries, transportation, telecommunication, etc. However, the development of these nonconventional energy resources in the Tanzanian environment is hindered by several challenges such as shortage of human capital and training, large investment cost, limited awareness, minimum and limited research and development, centralized approach (monopoly) regarding the institutional framework, and more interest in centralized electricity from the national grid (TANESCO) leading to minimal adoption of decentralized power schemes [10].

Biomass, in form of firewood or charcoal, is one of the nonconventional energy resources, which is used extensively worldwide, especially in developing countries for food preparation and heating purposes. Biomass is an organic material derived from animals and plants that is a renewable energy source [12]. It is made up of sun-stored energy. The energy from biomass is transformed into heat when it is burned. The process of photosynthesis, which occurs when trees and plants develop, takes energy from the sun to transform  $\text{CO}_2$  into carbohydrates such as starches, cellulose, and sugars [12]. Carbohydrates are organic molecules found in biomass. Once trees (plants) die, the energy stored in carbohydrates is released by the breakdown process and emits  $\text{CO}_2$  into the environment. Because the growth of new trees and plants replaces the supply, it is a renewable energy source. The use of renewable power is crucial because it promotes financial security, energy security, and ecological sustainability [12]. The transition from petroleum-based goods (i.e., fossil fuels) to substances derived from biomasses enhances the supply of long-term carbon-based feedstock. Some of the advantages of the biomass are as follows: (i) biomass energy is a renewable energy source that is derived from organic substances, plant, or animal waste, and these products are never depleted. Planting can be used to replace trees. (ii) It is healthier than fossil fuels; while burning, biomass emits  $\text{CO}_2$ , and it also collects  $\text{CO}_2$  for its own

development. However,  $\text{CO}_2$  emissions from fossil fuels are discharged into the air and are damaging to the ecosystem. (iii) Biomass energy is a cheap and widely available form of energy. Biomass can also be a long-term sustainable power source if trees are replanted. (iv) It minimizes dumping heaps of garbage (landfills). By burning it for energy, biomass converts environmentally hazardous waste into something beneficial and (v) reduced the requirement of fossil fuels. The use of biomass as an alternative fuel source decreases our dependence on nonrenewable energy sources, which is better for the environment and more cost-effective [12]. Some of the drawbacks of using biomass are (i) high investment cost. A boiler for biomass has a higher initial cost than a standard gas or oil boiler, and (ii) it requires huge space. Its boiler systems are typically bigger than oil or gas burners and need a fuel storage area, (iii) damaging the environment. Because biomass uses human and animal wastes, methane gas emissions are hazardous to people and the environment, and (iv) it may result in deforestation and (v) provision of inefficient energy. The performance of biomass fuel is inferior to that of fossil fuel [12]. When compared to fossil fuels, biodiesel derived from biomass, i.e., ethanol, is relatively inefficient. It can lead to extinction, habitat loss, fragmentation, and other negative consequences for biodiversity [12]. The drawbacks of the biomass utilization can be further expounded as follows: nonrenewable energy sources are good standard goods that can be easily integrated into a wide range of mature power conversion systems. Biomass, on the other hand, has lower energy content, owing to its higher oxygen content. Furthermore, because biomass is a relatively light and dense substance, the quantity of biomasses to be handled might be significant, posing storage and transportation challenges [13]. The usage of fuels with high moisture content reduces a power plant's total energy efficiency due to the need for power for hidden heat of vaporization. In addition to diminishing the overall heating value, high moisture causes a rise in transportation and storage costs, affecting the returns and the usability of biomass-driven power stations, particularly at higher power scales [13]. Biomass drying, on the other hand, provides little advantage for improving qualities such as oxygen concentration, poor grinding capability, and hygroscopic behaviour (absorption of moisture content) [13]. As a result, pre-treatment solutions must frequently be explored to improve the features of biomass as a fuel, making large-scale exploitation of these resources for bioenergy production uneconomic and inefficient [13].

The direct utilization of biomass-fired technologies presents several technological challenges, such as combustor fouling and corrosion induced by the alkalis of biomass wastes. Slagging and fouling impede heat transfer at combustor substrates and cause erosion and corrosion issues, reducing the equipment's lifetime [13].

Lignocellulosic biomass plays a great role in the biomass energy technology. Lignocellulosic biomass refers to the biomass originating from cellulose, which is cheap and easily accessible for producing energy [13, 14]. Forestry remnants, cellulosic wastes, agricultural remnants, and energy crops are all good sources of lignocellulosic biomass. The global

annual supply of lignocellulosic biomass is 181.5 billion tonnes [14]. The expense of this biomass is expected to range from US dollars 24 to US dollars 121 per tonne, type of crop, yield, area, and technique of analysis [13]. The total estimated energy yielded from the lignocellulosic biomass accounts for approximately 10% of world energy requirements [14]. Agricultural and forest leftovers alone contributed 30 EJ, a significant amount, to the yearly energy consumption of 4500 EJ [14]. There is biomass availability potential in Europe ranging from 615 to 728 million tonnes. In Italy, agricultural residues amount to approximately 12.8 million tonnes per year. 9.3 million tonnes can be generated from herbaceous crops (sunflower stalks, straw, and maize stalks) grown mostly in the north, and 3.5 million tonnes can be generated from arboreal crops (olive and fruit trees and vines) grown primarily in the south [13].

Tanzania possesses 33 million hectares of wood fields and forests, as per United Nations Food and Agriculture Organization's Global Forest Resources Assessment Study, but it loses approximately 400,000 hectares of forest each year [14]. Around 75% of harvested wood is not accounted for in government budget systems, resulting in revenue loss. The average household consumes approximately 47 kg of charcoal each month. Approximately 2500 million people worldwide rely on firewood for cooking and heating [10]. In the SSA region, including Tanzania, the population uses approximately 85 percent of all firewood that is taken from woodlands as energy or charcoal for cooking and heating [10]. For instance, as of 2018, the population was 56.32 million people. Only 35.6 percent of the overall population has electricity access; this corresponds to almost 36 million Tanzanians who do not have access to power. Biomass (e.g., firewood and charcoal) is still used as a source of energy by 80% of Tanzanians [10]. Whether burned inside or outside, the smoke created is a pollutant connected to negative health outcomes; particularly, respiratory disorders can drastically affect children and women [10]. Those who live in Tanzania's cooler southern highland regions frequently cook indoors, increasing their exposure to smoke pollution. Regardless of the high potential biomass in the country, there are only three power plants fuelled by biomass having at least installed capacity of 5.1 MW up to 18 MW, while the agro-industry-based power plants generate around 58 MW of their own electricity [10, 14].

Biofuels, principally bio-oil, bio-ethanol, bio-oil, and gasoline, and chemicals, can all be made from lignocellulosic resources [14]. For lignocellulosic biomass conversion, various conversion processes are available that take thermochemical, thermal, and biological routes [14]. The combustion of biomass for electricity and heat generation is referred to as thermal conversion. The process of converting biomass using extreme hotness and chemicals is termed as thermochemical transformation [14]. It is further classified into methods such as gasification and pyrolysis. These methods produce bio-oil and gasoline, as well as certain chemicals. Bioconversion is required for turning biomass into biofuels using microorganisms. Its primary product is bio-ethanol, although it also generates biobutanol, biogas (methane), and little chemicals [14].

Before bioconversion, pretreatment is essential to separate the ingredients of biomass for optimal fermentation [14]. Nonetheless, using feedstock necessitates a significant investment in power, heat, and work throughout the conversion process. The core of the problem is the costly pretreatment stage, which raises the entire cost of bioconversion. Chemical and physicochemical pretreatments are now the most effective. They are, however, harmful to the environment and produce hazardous by-products such as furfural. As a result, environmentally friendly biological pretreatment techniques are used on occasion, but with reduced fermentable sugar yields [14].

Integrated biorefineries increase the consumption and use of biomass to generate communal energy by mixing fundamental conversion, split-up, and downstream improvement processes. It is a necessity for fostering sustainable energy in a cost-effective manner. However, implementing them on a commercial basis is difficult. Recent research has focused on lignin chemistry and valorization, cost-cutting measures, sustainability of pretreatment procedures, advantages and disadvantages of various pretreatment methods, and transition processes of lignocellulosic biomass [14].

The conversion of lignocellulosic biomass into bioenergy is dependent on various processes, comprising (i) thermal transformation by burning, (ii) thermochemical transformation through both gasification and pyrolysis, and (iii) physiological (microbiological) transformation involving the use of biocatalyst [14].

*Thermal Conversion:* high-temperature direct combustion of lignocellulosic biomass produces thermal energy and ash particles. The thermal energy obtained can be used to generate electricity and power [14]. On a commercial basis, it is a low-cost method that has been thoroughly developed. However, it is not an environmentally favourable process because ash particles cause significant pollution [14]. Whole process is fully described as follows: direct combustion of biomass at hot temperatures (800–1600 C) produces thermal energy. When the fuel combines with oxygen, heat energy is produced, as well as flue gas containing CO<sub>2</sub> and water. The intensity of the heat flame can surpass 1650 C and is determined by the moisture contained in the biomass and heating value of the energy, the structure of the furnace, and the fuel-to-air ratio [14]. Because of their increased efficiency, biomass pellets are extensively used in commercial combustion in European countries. For this technique, biomass with moisture contained in biomass below 50% is recommended [14]. Combustion is carried out on a variety of scales utilizing a variety of equipment such as stoves, boiler, furnaces, turbo generators, and turbines. It is required for turning the chemical energy in biomass into mechanical energy. The combustion of fossil fuels results in massive emissions of CO<sub>2</sub>, NO<sub>x</sub>, PM (particulate matter), and cinders. These discharges to the environment represent significant greenhouse gas risks [14].

*Thermochemical Conversion:* the conversion of biomass into liquid or gas, which is then upgraded into biofuel, physical catalyst, or heat, is used. This method necessitates a high level of energy consumption. The chemicals utilized are



frequently costly, and as a consequence, inhibitors are produced. The cost of cleaning up emissions and pollutants is inextricably linked to the total price of the process [14].

*Gasification* is a thermochemical activity that combines the interaction of biomass in the presence of air, oxygen, or steam to generate a combination of gases, including hydrogen ( $H_2$ ), methane ( $CH_4$ ), carbon monoxide (CO), carbon dioxide ( $CO_2$ ), nitrogen ( $N_2$ ), and hydrocarbons called as syngas, producer gas, or synthetic gas, the quantities of which vary [14, 15]. The primary goal of the process of gasification is to maximize the yield of vapors while minimizing the production of unreacted char and hydrolysable hydrocarbons. There are basically three significant types of gasification processes: fixed bed, fluidized bed, and entrained flow [14]. Gasification generates heat that can be used by kilns or boilers. Because unattractive compounds such as tars are destroyed during gas combustion, the syngas used is barely cleansed. If the tar is not overwhelmed and the majority of the particulate matter is removed from the gasifier, syngas can also be used in combustion engines [14, 15]. Gas turbines encourage high-efficiency integrated gasification-combined cycle power, while good gas cleaning is required. Syngas is produced in a gasifier either explicitly or implicitly supplying temperatures of  $600^\circ C$  to  $1000^\circ C$  to the gasifier [14]. A successful gasification reaction has interval of time of around 3 to 4 seconds. Syngas is essential for the production of a wide range of chemicals and fuels such as organic acids, alcohol, esters, and hydrocarbon fuels [14]. Gasification has a broad range of applications being required for hydrogen production, thermally powered generation, and the synthesizing of chemicals and fuels [14]. This promotes energy industries based on gasification that generates an assortment of various chemicals and energy. Utilization of micro-turbine-driven biomass energy systems (syngas) and combined heat and power (CHP) gasification-based systems are examples of thermally powered generation [13].

*Pyrolysis*: pyrolysis is defined as a thermochemical process that can occur as an early step to combustion processes or as a standalone process for biofuel production. Pyrolysis is the degradation of lignocellulosic materials into an aromatic and carbon-rich solid in the lack of oxygen at temperatures starting from  $300$ – $900^\circ C$ . This procedure yields biochar (solid type), bio-oil (liquid type), and flue gases (gaseous condition) [14].

*Bioconversion* is the process of converting biomass into biofuel by employing biocatalysts either directly or indirectly as part of a prior pretreatment step [14, 15]. Complex carbohydrates are broken down into intermediate sugars, which are then fermented to make biofuels such as ethanol. Biocontrol agents, such as enzymes or a cluster of micro-organisms, enable this conversion. It is an extremely selective, high conversion efficiency method. Unfortunately, this procedure takes more time plus an extremely high pretreatment cost, indicating the process' infeasibility [14].

Direct microbial conversion (DMC) or consolidated bioprocessing (CBP) is a method of producing value-added products by combining enzymatic hydrolysis

and fermentation using a cluster or a sole organism. Without any pretreatment, lignocellulosic biomass is transformed straight into biofuels [14].

*Indirect Biological Conversion*: lignocellulosic biomass is first decomposed into fermentable sugars, which are then fermented into biofuels [14].

Geothermal energy is the energy (in form of heat) that is produced and retained in the earth. It is both clean and sustainable, and it can be found in the bedrock and fluids underneath the earth's crust. It is available from low ground to several miles beneath the surface and even deeper to the exceptionally hot molten rock known as magma [12].

Some merits of geothermal energy are as follows: (i) it is ecologically friendly: a geothermal power station has a very low carbon footprint. The  $CO_2$  emitted by geothermal power stations is one-eighth of that emitted by an equivalent coal power station. (ii) Stable resource: the plant's power output can be forecast and is not subjected to the equal low energy variations as solar or wind. (iii) A renewable resource: geothermal reservoirs are refilled naturally. (iv) No propellant involved: after construction, no mine or logistic activity is required. (v) Minimal land footprint: its plant installation requires a small amount of land. (vi) Economic criteria: some places are cost-competitive. (vii) Accessibility: most places have some amount of geothermal power available [12].

Some demerits of geothermal are as follows: (i) production of hazardous emissions: greenhouse gases (sulfur,  $CO_2$ , silica) are commonly found near plants. Toxic heavy metals such as mercury, arsenic, and boron may be present in the reservoirs. (ii) Geographical restrictions: suitable geothermal reserves are few. Prime locations are frequently located distant from major population centres. (iii) Pump powering costs: geothermal heat pumps require a power source. (iv) Expensive electricity: total costs for a megawatt geothermal power station typically range between \$ 2 and \$ 7 million. (v) Surface instability: the plant's construction may have an impact on the land's stability. (vi) Distribution costs: if energy is carried across great distances, the cost can become prohibitively expensive. (vii) Operation without a requirement of steam: if temperature is not properly managed, it can result in melting or other problems where energy is not dispersed or utilized effectively [12].

It has been expected that the global consumption of geothermal energy grows up to 200 TWh in the last two years (i.e., 2020) [16]. The annual growth rate of the consumed energy was estimated to be equivalent to 7%. The United States, Italy, Indonesia, Philippines, and New Zealand lead the world in geothermal power generation [10]. Tanzania has significant geothermal energy resources, but they have yet to be completely measured [10]. The assessment of geothermal energy resources began in 1976 and Tanzania now has a potential of around 5000 MW capacity that can be generated by geothermal power resources, with the majority of prospects situated in the East Rift Valley System in Africa [10]. Tanzania's geothermal blessings and prospects are divided into three major zones: the northeastern zone, which includes the some areas of Kilimanjaro, Arusha, and Mara;

the southwestern zone, which particularly includes the regions of Mbeya and Rukwa; and the eastern parts of coastal belt zone, which includes the Rufiji Basin and is associated with magmatic and rifting encroachments [10].

The intensity of the geothermal power source can range from 60°C to 350°C. Power stations are not cost-effective for temperatures under 80°C owing to the reduced overall system efficiency [14]. Temperature is used to categorize geothermal resources for different applications such as conventional electricity generation, binary cycle electricity generation, and heating (heat pump). The possible use of geothermal energy can vary depending on the level of temperature of the geothermal fluid, geological pattern, and hydrogeological features [17, 18]. Different geothermal technologies are used either directly or indirectly for different applications such as electric systems, district heating systems in cold zones, and cooling systems in hot zones [17–20]. Direct applications of heat derived from geothermal energy resource are beneficial to environment and are now replacing traditional sources of energy (fossil fuels) [18]. Possible applications of geothermal energy resources depending on the fluid temperature are such as electricity production at temperature above 90°C; direct Applications of thermal energy for industrial process, agricultural production, and space heating at temperature below 90°C; and exceptional operations at fluid temperatures lying between 30 and 50°C [17, 19]. Geothermal fields possessing low temperatures have been harnessed for industrial applications in China, Hungary, Iceland, Hungary, Germany, Russia, Turkey, France, and other nations for a number of decades, and many authors have recently proposed the use of geothermal energy alone or in conjunction (integrated) with several other nonconventional energy sources [19]. For example, in Kamchatka, Russia, geothermal field with low-temperature field of meteoric origin has harvested thermal water since 1966, mostly in the manner of artesian flow, to supply the multiple swimming pools, the district heating with two settlements, greenhouse gardening, and fish breeding [19]. Enhanced Geothermal Systems (EGSs) are man-made geothermal tanks formed by hydraulic fracturing and other stimulation techniques in ultralow permeability rocks. The principal purpose is to foster favourable circumstances for the circulation of water and heat mining in order to support surface energy uses such as district heating, power production, and straightforward industrial heat use [17, 20]. Technological advancements in hydraulic fracturing and multistage hydraulic fracturing have revolutionized flow permeability shale in recent years. Even though the high price remains a significant obstacle, it is hoped that this technology will be used to design fracture structures with a vast surface area of EGS (20). Since the 1970s, many efforts have been made to construct cost-effective EGSs. For example, the EGS at Fenton Hill in New Mexico was the first effort to harvest heat from porous rock. Multiple vertical hydraulic fissures were built to connect and allow circulation of water between two wells (holes) apart, extracting heat from 190 degree Celsius at 3500 metres depth. Projects of a similar nature have been carried out in Korea, Australia, and Europe. With hurdles from seismicity, connection, and financial feasibility, these have met with uneven success. Another recent project

is the thermal field known as Qiabuqia in China, which has been dug to a depth of 3.7 km depth [20].

There are numerous classifications for geothermal energy resources, based upon enthalpy or temperature (basic types are dry steam energy plants, flash steam energy plants, and binary ORC (organic Rankine cycle) or the concept of exergy) [14, 16]. It is primarily classified into three types based on the temperature or enthalpy, namely low, medium, and large enthalpy resources. The major vocabulary used for classification based on exergy is specific exergy rate (SER) or specified exergy index (SEI). Geothermal resources are divided into three categories based on SER or SEI: low, high, and intermediate exergy resources. Because it is not possible to determine the type of fluids based solely on temperature or enthalpy, SEI-based categorization of geothermal energy is also necessary [16]. For saturated steam and water, the SEI value ranges from 0 to 1 [14].

Geothermal resources can directly be utilized for heating purposes. Geothermal power can be used to generate power, desalinated water, and space cooling [17, 21]. Also, apart from the basic kinds of power generation via geothermal energy resource (i.e., dry steam energy plants, flash steam energy plants, and binary ORC) [14, 16, 21], it can also be integrated by other energy resources: *incorporation of geothermal power system in conjunction with other energy sources*: cascading refers to the pairing of various power-producing systems. Geothermal power nowadays is currently combined with solar, biomass, thermoelectric generators (TEGs), tri-generation, hydrogen production, and other kinds of direct use [21]. When geothermal resources are extracted, there is frequently residual enthalpy in the geo-fluid that might be utilized in a new application. A moderate temperature from geothermal deposit can produce power in an ORC power plant, where the heating rate of the geothermal saline is sufficiently high to be utilized in heating (heat pump) process [16].

*Geothermal and Solar Photovoltaic (PV) Hybrid Systems*: hybrid solar geothermal energy systems are characterized by their simplicity to set up in a variety of operating circumstances. They have been demonstrated to be reliable by employing low-cost collectors to set prices comparable with other kinds of electricity. There is also the possibility of multi-generation (heating/cooling) applications, as well as significant capital cost reductions through shared equipment within systems. The drawbacks of hybrid geothermal power systems include reduced efficiency cycles, rusting and scaling difficulties, and multi-generation facilities that necessitate numerous heat sinks [19, 21].

*Other Geothermal Energy-Based Hybrid Systems*: while the majority of research for geothermal-hybrid systems has focused on pairing geothermal and solar technologies, there has been significant research into systems that use geothermal technology in conjunction with organic material (biomass), hydrogen production, TEG, and cogeneration with additional freshwater production [16, 19].

*Improved Geothermal Systems:* drilling to a depth where the mantle's temperatures are adequately hot to transfer heat from the permeable rocks into a working fluid is required for enhanced geothermal systems (EGSs) [16, 20]. The fluid is then directly discharged, and energy is generated by any of the existing power plant types. Hot dry rock is relatively prevalent in the planet with average thermal gradients ranging from 25 to 30 °C/km deep. Some preliminary thinking suggests employing thick fluids to boost circulation pathways and lower rock permeability, allowing for smoother circulation starting from the injection well into the extraction well [16, 20].

Before drilling and designing geothermal plant, knowing scientific-technical data connected to the possibility of geothermal resources and the sustainability of utilization is a critical task for assessing the economic feasibility of geothermal exploitation and the creation of future project plants. More information such as the applicability of system periphery conditions during long-term utilization, the interference of fluid extraction and reinjection during production, and the efficacy of geothermal production systems can be gathered in this manner. Numerical simulation is recognized as a critical instrument for the development and evaluation of geothermal energy, which is otherwise regarded as highly dangerous [19].

As it has explained above, the development of renewable energy technology in Tanzania is at preliminary stage [5]. However, micro-hydropower and solar PV have been implemented in off-grid locations of the country [5]. Hydropower generation technology is mature, and due to its high potential of hydro-source, several projects of various scales are being implemented off-grid locations [11, 22, 23]. Geographically, Tanzania is located close to the equator, and therefore, it receives large amount of solar radiation in almost all parts of the country. The central region has highest potential of solar energy [24, 25]. The chief purpose of this work is to execute feasibility study and make validation by designing hybrid system using micro-hydropower and solar PV technologies. Therefore, the availability and high potential of hydro- and solar energy sources in Tanzania can be used to design decentralized electricity generating systems including optimal hybrid power plants for rural areas. Similar studies have been performed locally and internationally concerning optimization of hydro-solar PV hybrid electric generating systems for off-grid locations. Some of these studies have mainly used hydro-energy source from perennial rivers, water streams, and agricultural irrigation channels to conduct feasibility studies to assess the feasibility of generating electricity for off-grid locations [26–28]. Different research methods and techniques have been used for the simulation [29–31]. Kalla et al. proposed the hybrid micro-grid system by integrating micro-hydro-turbine system, wind, solar PV, and batteries using a 1~voltage source inverter. The target of the designed system was to balance the electrical energy source among the three sources using real-time approach termed as digital signal processor controller (DSPC). The results indicated that the grid voltage and frequency are maintained constant regardless of expected variations in loads and intermittency from the

penetration of wind and solar PV [32]. Li et al. optimized micro-hydropower plants/solar PV/pumped hydro-storage hybrid system and devised the day-ahead best performance strategy of the pumped hydro-energy storage (PHES) to balance the generated power curve and the consumed electrical load while maximizing the income of the storage. Optimization through chance constrained programming (CCP) and multiple objective PSO (MOPSO) was also done, and further the day-ahead optimal performance strategy, the pumped hydro-energy storage (PHES) balanced the generated power curve and consumed electrical load while maximizing the financial income of the storage. The authors investigated the performance of the model using meteorological traced past record of MHPS generation and consumed electrical power obtained from Sichuan Province in China. The simulation indicated that the technical and financial (income) implications of MHPS may be obtained on the Pareto front by fully considering the performance strategy and target. In addition, the similarity increase in generated power and load is about 7.89% under the proposed energy storage system [33]. Rajadurai et al. proposed SPV-micro for household application that involved an approach of replacing traditional storage equipment. This system operates the DC motor for pumping water to the water storage tank through the pipeline. When power is required, water is released directly to the sump into the MHT, which drives the generator. The authors have outlined the benefits of the proposed system, that is, inexhaustibility, eco-friendliness, and natural regeneration [34]. Syahputra and Soesant proposed micro-hydro/solar photovoltaic amalgamated system for rustic areas in Yogyakarta, Indonesia. The system was optimized using HOMER software and extended particle swarm optimization technique. Based on the optimization results, both micro-hydro from irrigation channel and solar PV energy sources have high potential for generating electricity. The system can generate power that meets energy requirements with surplus [35]. Poudel et al. developed stochastic models for economizing streamflow and empirical model of MHPS in Nepal. The models are integrated in the structure of hybrid 2 tool to simulate the long-term hourly operation of hybrid systems. The results have indicated that MHPS connected with half quantity of batteries in combination with a small pond of one-hour capacity of storing water is capable of supplying electricity sufficiently in compliance with stipulated energy by the Tier 3 level under tracking frame [36]. Das et al. designed the micro-hydro/solar PV/battery hybrid system to solve energy shortage in southern areas of Bangladesh. The system was evaluated by considering the potential of tidal energy at the Bay of Bengal. System optimization for technical and economic feasibility analysis was done using HOMER software. The cost-effectiveness of the system was dependent on location. In addition, comparison results between the proposed system versus diesel generator indicated that total NPC of the former is less by 27% and the estimated COE is about 0.21 \$/kWh. The COE for the hybrid is less than that of DG by 28% [37]. Clements et al. carried out the feasibility analysis of hydropower mini-grids for preparing meals using electricity as another alternative to



direct biomass-based cooking in ten families in Simli in rural community, Western Nepal. Data on cooking record and electric mini-grids were gathered, and their compatibility with the micro-hydropower grids and cooking practices is analysed. It was found that the operational cost for cooking using electricity was less expensive than the conventional wood-based cooking method. It was also suggested that the use of electricity for cooking in rural areas of Nepal is not affordable [38]. Hoseinzadeh et al. hybridized hydropower plant with solar PV and wind system for solving energy shortage of the power during the dry season in Iran. Mashhad Sarrood hydropower plant of 61 kW was used in the study. The optimal configuration was obtained using HOMER software. This was done by considering the fluctuations of the river flow and solar radiation. The findings indicated that the energy dearth improvement should be carried using suitable financial appraisal [39]. Das et al. optimized solar PV/biogas/pumped hydro-energy storage/battery bank hybrid system for radio transmitter station in India. The system performance was analysed using meta-heuristic optimization techniques such as water cycle algorithm and month of flame. The results of the aforementioned optimizations were compared with the findings obtained using the genetic method. The study intended to abate the NPC under the defined constraints. Sensitivity investigation was employed to evaluate the reliability of the system using loss of load probability. The study indicated that both suggested meta-heuristic approaches are suitable for acquiring finest optimal design. The optimized system configuration by water cycle algorithm obtained solar PV panel area equals 548.7 m<sup>2</sup> producing power capacity of 69.2 kW, biogas gas generator of 16 kW, battery banks of 21 units, and upper tank volume of 2081.5 cubic metres. The total NPC for the optimized system is \$ 0.813 mil. The system is cost-effective [40]. Patel and Chowdhury carried out the techno-economic feasibility analysis of micro-hydro with battery energy storage micro-grid system as a solution to rural electrification for rural areas in South Africa. HOMER Pro was used to optimize the system. The strength of the micro-hydro-resource on the price of electricity was examined. The findings of this study indicated that the stronger the resources the less expensive the system [41]. Zema et al. carried out the techno-economic feasibility analysis of micro-hydropower plants within irrigation systems in Calabria, Italy. The study proposed a simplified method of siting turbines by choosing their power and evaluating the charges and gains. Also, they provided suitable indicators for the design of MHPS in the existing irrigation systems. The results, through the case study, have highlighted the tiniest cost-effective turbine with power capacity of 5 kilowatts. High potential of MHPS from the existing system has much more gains on monetary investment than Italian market. More profits are particularly obtained during the rainy season than in the dry season in the average of 55% [42]. Halabi and co-researchers examined two distributed electric power stations in the area of Sabah in Malaysia. The arrangement of every station included dissimilar components such as solar PV, DG, power-converting devices, and battery banks. The authors confirmed the data, simulated various

scenarios such as decentralized DG, PV-DG battery, and PV battery for both stations, and compared the optimal designs based on the selected scenarios. They studied the performance according to the different infiltration levels to find out the effect of PV integration. The investigation was carried out using HOMER software. Technical, economic, and environmental viability was the criteria for comparison of the scenarios. The energy demand was sufficiently met at the least NPC and LCOE. Generally, the findings of the study indicated that the growth trend of energy production using sources of renewable energy was higher than standalone DG. In particular, PV-DG battery is more practically, economically, and ecologically sustainable than DG. In addition, PV battery was found to be the most nonpollutant system with greatest cost. It was observed that the existence of non-conventional power sources improves the operation of decentralized power systems and minimizes the requirement for energy storage mechanisms [43]. Hossain and his co-authors carried out the feasibility analysis of hybrid renewable energy to supply power to big tourist centre located in the South China Ocean in Malaysia. The area receives electric power supply from the decentralized DG. HOMER software was used as a research platform to perform the techno-economic analysis. The highest and average daily electrical load demand for the tourist centre was found to be 1,185 and 13,048 kilowatts, respectively. The finest optimal decentralized hybridized renewable energy-based system is build up by components such as solar PV, wind, DG, converter, and battery bank. The optimization results of NPC, LCOE, renewable fraction, and emissions of carbon dioxide (CO<sub>2</sub>) were found to be \$ 17.5, \$ 0.279/kWh, 41.6%, and 2,571,131 kg/year, respectively. The optimal results of NPC, LCOE, and DG system only were \$ 21.09 mil, \$ 0.343, and 5,432,244 Kg/year, respectively. The proposed hybrid renewable energy-based system is economically and environmentally better than decentralized DG system only [44]. Olatomiwa plus his fellow researchers studied the potentials of the solar and wind in the geographical and political areas of Nigeria. In the study, researchers used the meteorological data in the range starting from eighteen to thirty nine years. Also, the authors analysed the financial and technical viability evaluation of multiple energy source system, which includes solar PV, wind turbines, DG, and batteries. The system was anticipated to provide the electric power to the healthcare building and ensuring that the demand is met. HOMER software was used for carrying out the investigation. The observations of this study indicated that the Sokoto and Jos regions are located in high potential of wind energy, while all the six regions receive the sufficient levels of solar energy for various applications. Optimal results have indicated that the hybrid energy system is the finest choice for the all places. The first configuration of optimal hybrid energy is made up of solar PV-wind-DG batteries that are considered for supplying electricity to rural health centre applications at Enugu, Maiduguri, Sokoto, and Jos. The second one of the hybrid energy system is built by solar PV-DG batteries that have been considered the perfect option for supplying electricity to health facilities in distant places of Iseyin and Port Harcourt because of the abundance of

renewable energy sources. The third configuration of DG system only has highest LCOE and CO<sub>2</sub> emissions of \$ 0.911/unit (kWh) and 9211 Kg/year, respectively, in all places under study [45]. Olatomiwa implemented the analysis of decentralized hybrid renewable energy-based systems to supply electricity to the health facilities in the off-grid rural areas or in the areas receiving unreliable power from the grid in Nigeria. These hybrid energy systems intended to address the problem of excessive motherly and child death rates in the country. The researchers described the statistical investigation of solar and wind energy source potentials for some particular remote areas according to the long period of time (in hours) and daily climatological information. The potentials of the energy sources have been validated by the techno-economic analysis of designing and sizing the solar PV-wind battery using HOMER software. The findings of this research indicate that Sokoto and Jos are located in high wind regimes for large power wind systems, while the rest sites are appropriate for small wind applications. On the other side, all sites receive sufficient coverage of solar energy for solar applications. The optimized solar PV-wind-DG battery hybrid system for rural health care is suitable for Sokoto, Iseyin, Maiduguri, Enugu, and Jos. The optimal PV-diesel battery hybrid system is suitable for Port Harcourt due to the available high potential of renewable energy sources. The authors concluded that the high potential of wind and solar sources in the nation makes a best situation for including the renewable energy sources in the planning and execution of the decentralized energy systems in the country for improving the provision of rural health care [46].

The novelty of this study is to explore the feasibility of generating electricity using abandoned potential of non-perennial tributary particularly in the areas of prolonged period of high rainfall (installation of small ROR power station hybridized with solar energy plus batteries). Also, this study presents the utilization of traditional approach of measurements to acquire the input parameters of the tributary. The objective of this study was to optimize the renewable energy-based hybrid system to generate electricity for the selected site. This study explores the neglected hydro-energy potential of a nonperennial tributary incorporated with the availability of abundant solar energy source. The tributary is deemed to have considerable untapped high potential of generating electricity for rural communities in the regions of prolonged high rainfall of Tanzania. This research work tries to explore more diversification of local renewable energy sources. In the study, the estimated water potential derives from abandoned potential of existing tributaries in the country including the Ikata tributary, which was designated as a case study for this research. The tributary is elevated and has suitable water head and sufficient water flow during the rainy season that may be adequate for supporting the rural electrification in the nation. Unfortunately, this potential of the tributary remains untapped. In the rainy season, this tributary has large amount of water low, which is simply flowing down into the permanent river called the River Shongo. This river provides water mainly for drinking and small agricultural irrigation in the community. The study is anticipated to create awareness

about how the kinetic energy of water from the disused potential of nonperennial tributary in zones of plentiful precipitation and solar energy source through the hybridization of the two renewable energy sources and battery energy can be harnessed. The study has also taken into account the benefit of high solar radiation availability. Tanzania is located in proximity to the equator, thus receiving substantial amount of solar energy. The potential of water source from a nonperennial tributary and solar energy source is therefore combined into hybrid power generation system. The hybrid system is optimized by the minimization of the costs such as total NPC and COE.

Secondly, in many feasibility studies that have been carried out, there is a limited use of traditional methods for data gathering on hydro-energy source. Traditional methods for the measurement of parameters of water flow rate and head refer to methods in contrary to the advanced ones. They are manually operated and use simple materials and tools for measuring the aforementioned parameters. Traditional methods are bucket methods, float method, and velocity area method, while approaches used for measuring head are, for example, dumpy levels and theodolite. These methods are simple and use simple local materials and inexpensive measuring tools. However, traditional methods need more number of persons to perform the task for measurements and can get hindrance during bad conditions such as massive rain and floods. They have also limitations in terms of accuracy and suitability for only small magnitude of water flow rate [47, 48]. These traditional methods may be supportive in developing countries where poor economy is a major challenge. In this research work, the techno-economic feasibility analysis is performed for increasing the access to electricity to small communities located at Ikukwa Village in Mbeya Rural District, Tanzania. This place consists of nonperennial tributary called Ikata, and its water flow is running down from the hills surrounding the peak of Mount Mbeya. In this feasibility study, case study is carried out at Ikata tributary using its nonperennial hydro-energy source to optimize hybrid micro-hydro-solar PV. Hybrid Optimization of Multiple Energy Resources (HOMER) Pro software is used for the techno-economic analysis for obtaining the optimal configuration of the proposed hybrid system. The analysis of this study has been presented in two stages. In the first stage, the Hybrid Optimization of Multiple Energy Resources (HOMER) software has been used to define the best possible configuration at the lowest total NPC plus COE.

Nowadays, many researches are being carried out using various artificial intelligent (AI) methods. Among these, AI methods include the newly invented artificial intelligent (AI) optimization approaches called PSO, GWO, and GWO-PSOHD. The major merits of the PSO are uncomplicatedness, easiness of procedure, great merging degree, less requirement of storage, and reduced reliance on the preliminary points. The GWO algorithm consists of features such as great convergence degree, performance independence, and heftiness [49]. The GWO-PSOHD optimization method is also lately established of which GWO is combined with PSO for improving the working ability of GWO. The GWO-PSOHD technique has been verified of being simple,



effective, easy, and competitive. Therefore, by employing the above named AI approaches further optimization results have been found out. The AI algorithms have been implemented in the MATLAB program. Through the optimization results, which are obtained from the HOMER software and AI methods, comparative study has been made for validation purposes. The AI optimization techniques provide lower values of total NPC and COE of the proposed hybrid energy system. In summary, this study contributes to the further exploration of unexploited potential in order to diversify renewable energy sources for rural electrification. Moreover, in the paper the estimated capacity of the tributary under study has been estimated through the traditional approach, which is rarely used. Also, the paper involves applications of AI techniques such as PSO, GWO, and GWO-PSOHD algorithms for sizing off-grid system, which further reduces NPC and COE. Similarly, in the paper the system is optimized by taking into account the system reliability.

The structure of this study is as follows: generating electricity through hybrid renewable energy-based system is presented in Section 2. Energy data collection and analysis are provided in Section 3. Electrical load demand is explained in Section 4. The proposed hybrid renewable energy-based system is described in Section 5. The results for technical and economic analysis obtained by HOMER software is provided in Section 6. PSO, GWO, and GWO-PSOHD methods (AI optimization algorithms) are explained in Section 7. Optimization results of technical and economic analysis obtained by GWO technique are delineated in Section 7. Recommendation for the long-term sustainability of the tributary under study is presented in Section 8. Future research work is offered in Section 10.

## 2. General Description of Electricity Generation through Hybrid Renewable Energy-Based System

There is abundance of electricity in Tanzania though this high potential has not been fully utilized to meet electrical demand, particularly in the rural and isolated areas. Table 1 indicates Tanzania's potential of renewable energy sources [5]. Table 1 indicates Tanzania's potential of renewable energy sources.

Nowadays, the URT immensely supports renewable energy resources for electricity generations. It has been projected that 100% of the total generated electricity in the country in the year of 2050 will be produced from renewable energy sources. In this projection, total power capacity will be approximately 60 gig watts [5]. Hydro- and solar energy sources are not only technologies but also preferable. Power generated from hydro-energy source is reliable and affordable. Solar energy-based systems have been installed in homes, health centres, dispensaries, and schools. Descriptions for hydro- and solar energy sources are given as follows.

**2.1. Hydro-Energy Resource.** Generating electricity through hybrid renewable energy system is a combination of different but complementary sources of energy. Hybrid

technology is reliable and economical. Hybrid power generation configuration is basically subdivided into three categories such as series, parallel, and switched hybrid configuration. Of these three categories, parallel hybrid configuration is the most preferable because it is reliable and energy source within it operates independently. Micro-hydro- and solar PV electric systems are dirt-free systems. Micro-hydro-generating system changes the dynamic energy of water into mechanical power by means of turbine, which is then converted to electricity by generator. Micro-hydropower (MHPS) according to the generating capacity refers to system with a range of capacities starting from 5 kW to 100 kW. Besides, the capacity for MHPS may reach up to 200 kW. Power output generated by MHPS depends on water head and flow rate. This power can be estimated as follows [28]:

$$P = \rho \times Q \times g \times H, \quad (1)$$

where  $P$  = power output produced by MHPS in watts (kW),  $\rho$  = water density in Kg/m<sup>3</sup>,  $Q$  = water flow rate in m<sup>3</sup>/s, and  $H$  = water head in  $m$ . MHSP can generate power for enhancing socioeconomic activities in the rural societies.

Most of the hydropower plants are classified ROR hydropower plants. The power production of ROR plants is not fixed and naturally fluctuates daily, weekly, and seasonally in river discharge. ROR plants can only generate electricity if sufficient amount of water flow of the river is available. If the river discharges fall below the minimum technical inflows, electricity generation ceases. It is worth noting that ROR hydropower projects with pondage as opposed to those with pondage can store water for greatest power consumption and continuously for base load, especially during the rainy periods. Advantages of the ROR power plants without pondage/reservoir for energy production are less impact on the available ecosystem; design is less complicated; overall lower capital costs; smaller water heads; limited time of construction; and offers the possibility of decentralized electrification, relatively low operational and maintenance cost. The design and operation are a challenging task in terms of robustness, flexibility, and periodic fluctuations.

**2.2. Solar Energy Resource.** Tanzania is a tropical country with a high potential of solar energy within a range of 4–7 kWh/m<sup>2</sup>/day [25]. Regardless of this sufficient amount of solar energy, there is a limited installed capacity of decentralized systems in rural areas. A solar energy source is an intermittent in nature causing power output variability. Power output depends on the amount of solar radiation, orientation, temperature, and site conditions [47, 50]. A battery energy storage system is required to store energy during high solar radiation and supply electrical load whenever solar radiation is poor. In a hybrid renewable power generation system, energy storage cost is reduced and its performance is also reliable [47, 48]. Electricity generation by solar PV technology consists of solar panels, charge controllers, deep cycle

TABLE 1: Tanzania's potential of renewable energy sources.

| Renewable energy source | Approximated potential                                   | Growth in percentage         |
|-------------------------|----------------------------------------------------------|------------------------------|
| Mega hydro-energy       | 4.7 GW                                                   | 12%                          |
| Small hydro-energy      | 485 GW                                                   | 2%                           |
| Solar energy            | Greater capacity than 200 $W_p/m^2$                      | Solar radiation: 215 $W/m^2$ |
| Biomass energy          | Greater capacity than 500 MW                             | 34 million $m^3$ /year       |
| Geothermal energy       | 5000 MW                                                  | —                            |
| Biofuels                | Existence of plants for ethanol and biodiesel production | —                            |
| Wind energy             | Greater speed than 8 m/s                                 | —                            |
| Tidal and wave energy   | Ongoing assessment                                       | —                            |

batteries, and power converters. Power output is calculated as follows [28]:

$$P_{\text{Peak}} = \frac{E_{el} \times I_{STC}}{E_{\text{global}} \times PF}, \quad (2)$$

where  $P_{\text{peak}}$  = peak power (KW p) under standard test conditions (STCs),  $E_{el}$  = actual annual energy output (kWh/annual),  $I_{STC}$  = irradiance solar radiation ( $1 \text{ kWh}/m^2$ ) under STC,  $E_{\text{global}}$  = annual global solar radiation ( $\text{kWh}/m^2/\text{year}$ ), and PF = performance factor.

### 3. Energy Data Collection and Analysis

Based on the novelty and proposed hybrid MHPS-solar PV, a case study is made at the location where there is a high potential for a nonperennial tributary in a prolonged rainfall including the high potential of solar energy. The following sections provide a brief description of area of study and methodology of this study.

**3.1. Description of Area of Study.** Ikukwa Village is one of the villages in Mbeya Rural District of Mbeya Region (latitude  $8^\circ 42.6' \text{ S}$ , longitude  $33^\circ 27.9' \text{ E}$ ), Tanzania. The village comprises 14 sub-villages with 570 total households. Geographically, and logistically, the village is located beyond Mount Mbeya where it has been difficult for direct grid extension from Mbeya City. The location of this study is latitude  $8^\circ 45.4' \text{ south}$  and  $33^\circ 17.4' \text{ east}$ , as indicated in Figure 1. This research is carried out in three main phases such as determination of theoretical capacities of solar photovoltaic system (SPVS) and MHPS, the analysis of energy demand of Ikukwa Village, and the optimization of proposed hybrid system. Hybrid optimization multiple electric resources (HOMER) Pro software is used for simulation during the techno-economic analysis of the hybrid energy micro-grid system.

Logistically, during the rainy season the road to the area of study is also impassable. Recently, only 40 households of the total number of households of the village are connected to the national grid distributed by Tanzania Electricity Supply Company (TANESCO). The availability of grid connectivity in the whole village is equivalent to 7%. One of the reasons is that this village has dispersed households and poor population density. Table 2 indicates the status of grid connectivity percentages by sub-villages. This village consists of tributary called Ikata. The Ikata tributary is a

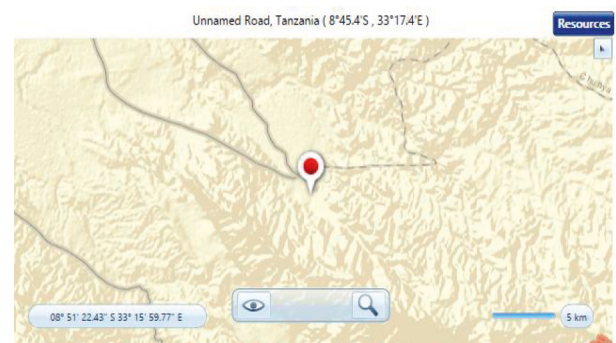


FIGURE 1: Location of Ikukwa Village in rural areas Mbeya Region, Tanzania.

TABLE 2: Status of national grid connectivity (TANESCO) at Ikukwa Village (source: survey by authors before 31.07.2019).

| Subvillage  | NH  | Grid-connected customers | % of grid-connected NH |
|-------------|-----|--------------------------|------------------------|
| Mdonya      | 26  | 0                        | 0                      |
| Kiwanja     | 38  | 0                        | 0                      |
| Unguja      | 28  | 0                        | 0                      |
| Shongo      | 33  | 01                       | 0.2                    |
| Juakali     | 42  | 0                        | 0                      |
| Ujamaa      | 34  | 01                       | 0.2                    |
| Kariakoo    | 41  | 0                        | 0                      |
| Ukwaheri    | 29  | 0                        | 0                      |
| Ikukwa      | 52  | 14                       | 2.5                    |
| Mbuwi       | 42  | 0                        | 0                      |
| Zongo       | 59  | 08                       | 1.4                    |
| Itende kati | 90  | 10                       | 1.8                    |
| Mahonza     | 28  | 05                       | 0.9                    |
| Itende juu  | 28  | 01                       | 0.2                    |
| Total       | 570 | 40                       | Approx.7. %            |

NH = number of households.

nonperennial water stream having a high potential of generating electric power during a rainy season. This stream of water flows down from the hills surrounding Mount Mbeya to the River Shongo. Figure 2 indicates the Shongo River, which receives water supply from Ikata tributary at the village. Other tributaries that deliver water supply to River Shongo are Ho, Ilwilo and Igodela. The perennial River Shongo supports Ikukwa dwellers by providing them with water supply for drinking and irrigation for agricultural activities by means of small channels. According to the fieldwork carried out by the authors, Ikata tributary is more



FIGURE 2: Shongo River receives water supply from Ikata tributary at Ikukwa Village from the hills of Mt. Mbeya, Mbeya Region, Tanzania (photo during the dry season).

preferable for distributed electric generation than the Shongo River because the former has a substantial water head and is closer to the proposed site of study. It is well known that distributed generation is installed close to the electrical load for improved reliability and efficiency. For these reasons, Ikata tributary is considered to have a high potential of generating electricity when hybridized with solar PV-generating system. Secondly, Mbeya Region where Ikata tributary is found generally has high rainfall for a prolonged period of time for sufficient water flow rate during the rainy season. The hybrid of MHPS of run-over-river category and solar PV-generating system is intended to increase power accessibility to sub-villages of Ikukwa Village, useful for reducing the challenge of frequent and longstanding load shedding from the national grid network, and can support productive activities in the Ikukwa community.

The steps for this research activity are presented in Figure 3. The steps involved are as follows: this research process follows the steps such as literature review, field appraisal, and energy assessment for the proposed hybrid system including gathering data for miscellaneous meteorological information, energy resource, and demand assessments, and determination of the optimal hybrid renewable energy systems based on the lowest total NPC and COE, judgement is made about whether the analysis gives optimal results or not. If yes, optimal results are presented, and finally, conclusions are made. In contrast, the analysis is repeated until optimal results are achieved.

**3.2. Methodology.** Collected data for hydro-potential of MHPS were achieved from a field appraisal in the proposed site. The hydro-energy assessment is concentrated on the tributary of Ikata at Ikukwa Village, Mbeya Rural District, Mbeya Region, Tanzania. In this tributary, direct measurements for the estimation of water flow rate and head parameters were obtained through traditional methods. Moreover, the potential for solar radiation and other meteorological information was achieved from the databank of the National Aeronautics and Space Administration (NASA) [51, 52] and Tanzania Meteorological Authority (TMA) via Mbeya Meteorological Station (MMS), respectively. Miscellaneous meteorological data are important for

understanding how the changes in temperature and rainfall affect the performance of MHPS and the proposed hybrid energy system in terms of power output. Table 2 indicates the status of national grid connectivity (TANESCO) at Ikukwa Village (source: survey by authors before 31.07.2019) (Table 3).

**3.3. Platform Used.** HOMER software is micro-grid modelling software. It was established by National Renewable Energy Lab (NREL) under the department of energy in the United States (USA). HOMER is popular leading software used for the optimization of micro-grid system. It is commonly used for feasibility studies to determine the optimal configuration of the designed system. The factor that is used for the selection of the optimal configuration in the simulation is the total lowest net present cost (NPC) and cost of energy (COE) [53].

In addition to HOMER software, the GWO algorithm is used for the technical and economic analysis of the system by incorporating the system reliability constraint. In this specific study, loss of electrical power probability (LEPP) has been taken into account as reliability restriction. The optimum system is obtained at the lowest total NPC (also the lowest COE) under reliability boundaries. The detailed description of the technical and economic parameters used in the analysis is as follows [54]:

$$LEPP = \frac{\sum_{t=1}^T SPS(t)}{\sum_{t=1}^T P_L(t)}, \quad (3)$$

where SPS is the shortage in power supply,  $P_L$  is the total electrical load, and  $T$  is a period of time for a year, which is equivalent to 8760 hours. If LEPP equals zero, it implies that the hybrid micro-grid system produces sufficient power for the load, and if the LEPP is 1, it means that the system is incapable of satisfying the load [40].

$$SPS = P_L - (P_{\text{ghybrid}} + SOC_t - SOC_{\text{mini}}), \quad (4)$$

where  $P_{\text{ghybrid}}$  is the total power generated from the hybrid micro-grid system,  $SOC_t$  is the state of charge of the battery at time  $t$ , and  $SOC_{\text{mini}}$  is the minimum state of charge of the battery.



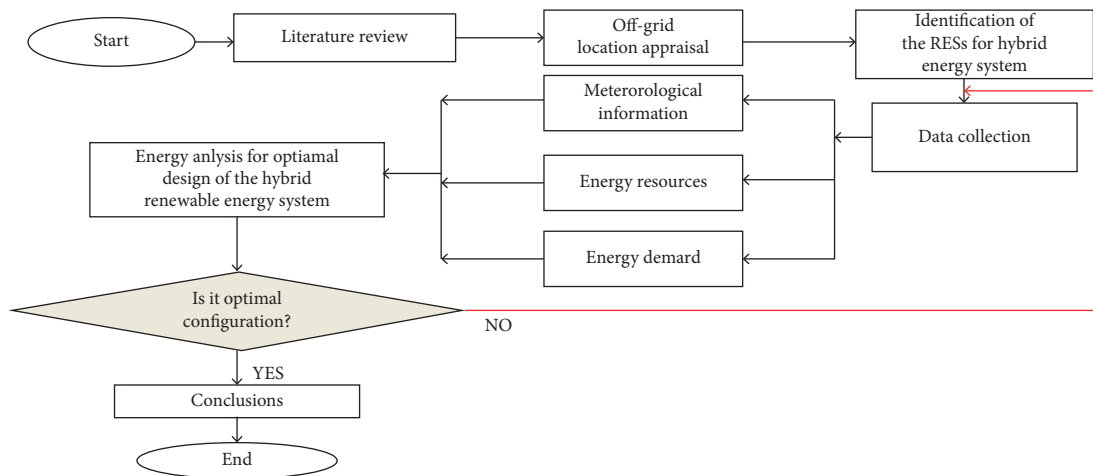


FIGURE 3: Steps for implementing the research.

TABLE 3: Total electrical load at Ikukwa Village (7% access to grid from TANESCO) (source: survey by authors before 31.07.2019).

| Name of small communities of Ikukwa Village | Estimated load in watts (W) |
|---------------------------------------------|-----------------------------|
| Ujamaa                                      | 115 803                     |
| Ikukwa                                      | 149 531                     |
| Jua kali                                    | 141 482                     |
| Kiwanja                                     | 52 306                      |
| Ukwaheri                                    | 46 079                      |
| Itende juu                                  | 33 063                      |
| Mbuwi                                       | 80 626                      |
| Mahonza                                     | 49 799                      |
| Kariakoo                                    | 79 475                      |
| Unguja                                      | 33 404                      |
| Mdonya                                      | 61 372                      |
| Zongo                                       | 100 825                     |
| Shongo                                      | 42 448                      |
| Itende kati                                 | 129 953                     |
| Total                                       | 1116166                     |

TANC comprises all annualized costs of the system in \$/year. It consists of annualized investment charge (AIC), annualized spare charge (ASC), annualized fuel charge

(AFC) of energy source, and annualized operation and repair charge (AOMC) of the hybrid system. TANC is computed in the following equation [40]:

$$TANC = ACC (MHPS + Solar PV + Batteries + Converter) + ASC (MHPS + Solar PV + Batteries + Converter) + AFC (MHPS + Solar PV) + AOMC (MHPS + Solar PV + Batteries + Converter). \quad (5)$$

However, in this study AFC is neglected as hydro- and solar energy sources are free of charge, and for simplicity, salvage value is not considered.

The NPC can be simply defined as a sum of whole expenses of project's lifespan as follows [40]:

$$NPC = \frac{TANC}{CRF(i, Pproj)}, \quad (6)$$

where  $i$  indicates the interest rate in %,  $Pproj$  is the lifespan of the project in years, and  $CRF$  represents the capital recovery factor.

COE (\$/kWh) refers to the average charge per unit of useful generated electricity. COE can be computed as follows [40]:

$$COE = \frac{(NPC * CRF)}{EL}, \quad (7)$$

where  $EL$  is the total annual energy consumption (kWh/year). COE is alternatively computed in terms of TANC by the following equation [40]:

$$COE = \frac{TANC}{EL}. \quad (8)$$

CRF is defined as follows [40]:

$$CRF = \frac{i_r * (1 + i)^N}{(1 + i)^N - 1}, \quad (9)$$

where CRF is defined as the capital recovery factor,  $i_r$  equals the interest rate, and  $N$  is the lifespan of a hybrid micro-hydro/solar photovoltaic system with battery energy storage.  $i_r$  is defined as follows [40]:

$$i_r = \frac{(i - \text{inflation rate})}{(1 + \text{inflation rate})}. \quad (10)$$

#### 4. Assessment of Energy Resources, Meteorological Information, and Electrical Load Demand

The main goal of this research is to optimize the hybrid for the same mentioned technologies. A hydro-energy source originates from a nonperennial Ikata tributary. Hydropower technology requires two major parameters for power generation to meet forecasted demand. These required parameters are suitable to head and water flow rate. The assessment of the parameters needs deep analysis of a number of facts regarding water stream such as rainfall, nature of ground, and vegetation at certain period of time. There are a number of methods, and both traditional and modern methods are employed for determining the abovementioned parameters. However, traditional methods are useful for the initial estimation of energy potential, especially in a limited time, and are mostly applicable in developing countries. Therefore, data for water head and water flow rate were collected by traditional methods. Data for solar radiation were obtained from the NASA-SSE, and energy demand was collected from the inventory of appliances.

**4.1. Assessment of Hydro-Energy Potential of Ikata Tributary.** Part (a): assessment of hydro-energy resource for MHPS using the traditional approach.

Water flow rate and head are required input parameters to HOMER Pro software and are determined using traditional methods as follows [55].

**4.1.1. Materials and Methods for the Determination of Water Flow Rate.** The estimation of water flow rate float method is used. This method requires the determination of surface water velocity, width, and depth of water stream. Finally, these dimensions enable the calculation of water flow rate.

**(1) Estimation of Average Velocity of Water Stream.** Cylindrical wooden coloured floats of individual diameter and length of 5 cm and 30 cm, respectively, are used to assess water surface velocity of River Ikata. The distance of 10 cm between two marks by means of two pieces of bamboo tree was measured using a measuring tape. Stopwatch was used to monitor time in seconds for float movement over marked

distance. Ten-time measurements were recorded and lastly averaged. Table 4 shows time measurements.

Sum = 16 + 16 + 16 + 15 + 16 + 15 + 16 + 17 + 16 + 16 = 159.

Therefore, the average time for the movement of float at the measured distance is 15.9 s. Surface water speed equals the distance along the water stream per average time, which is 10 m/15.9 = 0.63 m/s.

Average water velocity  $V$  = surface water speed  $\times$  correction factor = 0.63  $\times$  0.85 = 0.536 m/s.

**(2) Estimation of Average Width of Water Stream.** The average depth of water stream was measured using the traditional method by means of pieces of bamboo tree and measuring tape. Pieces of bamboo tree were laid at equal spaces across water stream, and the distance between two ends was marked. At the end of this process, pieces of bamboo trees were taken aside, and from two marked ends of individual piece of bamboo tree, distance, which is equivalent to water stream, was measured by a measuring tape. Table 5 indicates width measurements.

As 6.1 m has the highest frequency among other width measurements, therefore this value is used as the average width of water stream of Ikata River. Therefore,  $W = 6.1$  m.

**(3) Estimation of Average Depth of Water Stream.** The depth of water stream was measured using pieces of bamboo tree and measuring tape. Several pieces of bamboo tree were dipped vertically to the bottom of the water stream along the width of water stream, and the water level of each bamboo stick was marked using marker pens. These sticks/rods were taken out for measuring distance starting from the marked point of individual piece to the bottom. Table 6 shows depth measurements.

The deepest measurement from the table above is 1.8 m. The estimation for average depth of water stream is half of the deepest measurement; that is,  $d = 1.8/2 = 0.9$ .

**(4) Determination of Water Flow Rate.** Having estimated average water velocity, average width and average width water flow rate can be found. A water flow rate of the river is the product of three above named parameter. Therefore, water flow rate =  $V \times w \times d = 0.536 \text{ m/s} \times 6.1 \text{ m} \times 1.3 \text{ m} = 2.943 \text{ m}^3/\text{s}$ .

**4.1.2. Materials and Methods for the Determination of Water Head.** Water head is determined through traditional or local methods known as the carpentry method. Materials and equipment required are measuring tape, vertically erected wooden piece, that is, pieces of bamboo tree (naturally marked), and horizontal rectangular board and mason's level. A piece of chalk or marker pen has been used for marking the height of the bamboo during the measurement of water head using the carpentry method. Measuring measurements are used to take measurements for distance; pieces of bamboo are vertically erected to measure height; horizontal board and mason's level are used for horizontal levelling from reference on the ground to required height on a vertical piece of bamboo (marked by chalk or marker pen).

TABLE 4: Time measurements.

| Measurement | Time (s) |
|-------------|----------|
| 1           | 16       |
| 2           | 16       |
| 3           | 16       |
| 4           | 15       |
| 5           | 16       |
| 6           | 15       |
| 7           | 16       |
| 8           | 17       |
| 9           | 16       |
| 10          | 16       |

TABLE 5: Width measurements.

| Measurement | Width (m) |
|-------------|-----------|
| 1           | 6.7       |
| 2           | 6.1       |
| 3           | 6.1       |
| 4           | 6.1       |
| 5           | 7.0       |
| 6           | 6.8       |
| 7           | 6.9       |
| 8           | 6.6       |
| 9           | 6.0       |
| 10          | 5.6       |

TABLE 6: Depth measurements.

| Measurement | Depth (m) |
|-------------|-----------|
| 1           | 1.6       |
| 2           | 1.6       |
| 3           | 1.5       |
| 4           | 1.3       |
| 5           | 1.7       |
| 6           | 1.8       |
| 7           | 1.4       |
| 8           | 1.2       |
| 9           | 1.4       |
| 10          | 1.3       |

Before estimating water head, intake position and location of turbine are identified. Table 7 displays water head measurements.

**4.1.3. Calculation of Hydropower Capacity.** Theoretical power potential depends on two major factors, namely flow rate and head of the falling water. The potential of micro-energy resource is evaluated using (1) as follows:

$$\begin{aligned}
 \rho &= 1000 \text{ Kg/m}^3, \\
 g &= 9.8 \text{ m/s}^2, \\
 Q &= 2.943 \text{ m}^3/\text{s}, \\
 H &= 13 \text{ m}.
 \end{aligned} \tag{11}$$

Therefore,

TABLE 7: Water head measurements.

| Measurement | 1   | 2   | 3    | 4   | Gross head in hills (m) |
|-------------|-----|-----|------|-----|-------------------------|
| Height (m)  | 3.4 | 3.4 | 4.00 | 2.6 | 13                      |

$$\begin{aligned}
 P &= 1000 \times 9.81 \times 2.943 \times 13 = 375,320.79 \text{ W}, \\
 P &= 375 \text{ kW}.
 \end{aligned} \tag{12}$$

This is the theoretical maximum power capacity. This capacity is expected in the month of January due to the highest rainfall in the location of the site project.

Part (b): assessment of hydro-energy resource using HOMER Pro software.

Depending on the surveyed parameters of Ikata tributary implemented in HOMER Pro software, Figure 4 indicates the annual monthly hydro-flow rate. The scaled annual average water flow rate is 899 litres per second. The comparison of the two figures, namely Figures 4 and 5, indicates the fluctuation of rainfall in a year and it affects the water flow rate of the tributary under study, hence the power output from MHPS within the proposed hybrid technology. The drastic effect of poor rainfall on water flow in the tributary is overcome by the integration of SPVS with batteries. The focus of this research is to investigate the power generation likelihood of the tributary and the general functioning of the proposed hybrid energy micro-grid system.

**4.2. Solar Energy Resource.** Data for monthly solar radiation are obtained from the database of NASA [51]. Mbeya Region is located at latitude  $8^\circ 42.6'$  south and longitude  $33^\circ 27.9'$  east. Data for monthly mean solar global horizontal irradiance in terms of solar radiation and clearness index are presented in Table 8 [51]. Data on solar energy for Mbeya Region are used for a selected study area in Mbeya Rural District. The annual average of solar radiation analysed by HOMER software for a proposed project site is  $6.11 \text{ kWh/m}^2/\text{day}$ . The potential of solar energy for the site and global solar resource are given in Figures 6 and 7, respectively.

**4.3. Miscellaneous Meteorological Information.** Meteorological information is obtained from Tanzania Meteorological Authority (TMA) at the branch of Mbeya Meteorological Station (MMS) [52]. The required meteorological data for this study are rainfall and temperature. In the month of January, the area of study has the highest average rainfall about 247.78 mm. The average rainfall for May and October is 11.60 mm and 16.70 mm, respectively. In the period from June to September, the rainfall is zero. The amount of rainfall in the months of May and October is low. However, water flow from the tributary under study in such months though is assumed to be sufficient to generate power depending on geological conditions. Rainfall has a large impact directly on the amount of water flow rate of the nonperennial tributary and indirectly on the operation of MHPS in the proposed hybrid energy system. Therefore, no power will be produced from MHPS. Instead, SPVS with battery energy storage will produce power within the hybrid



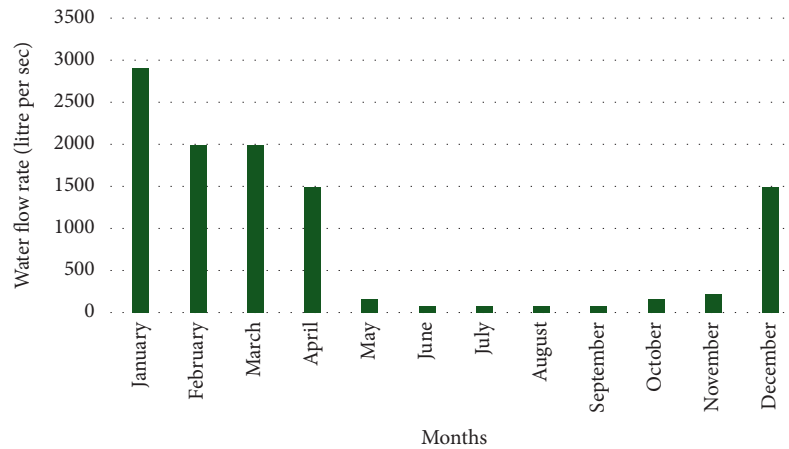


FIGURE 4: Annual monthly average water flow rate.

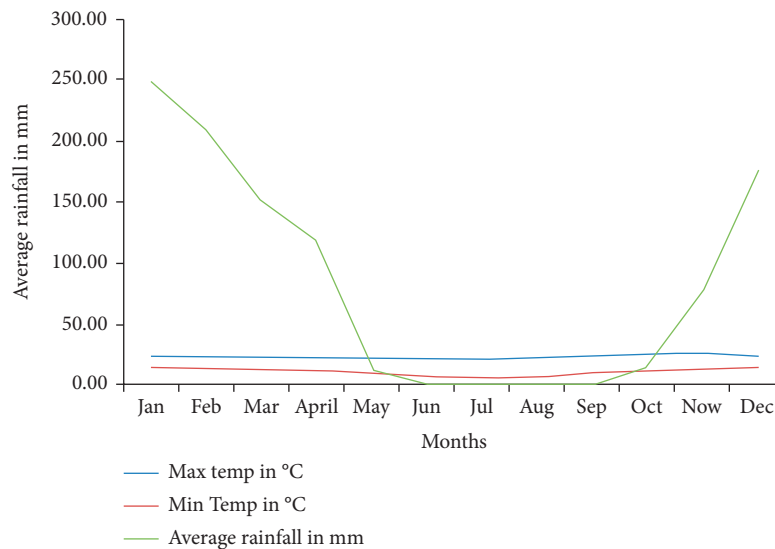


FIGURE 5: Rainfall and temperature for the location of the proposed project in Mbeya Region, Tanzania (2000–2015) [52].

TABLE 8: Data for monthly solar radiation and clearness index [51].

| Month     | Clearness index | Solar energy radiation kWh/m <sup>2</sup> /day |
|-----------|-----------------|------------------------------------------------|
| January   | 0.48            | 5.27                                           |
| February  | 0.51            | 5.54                                           |
| March     | 0.55            | 5.83                                           |
| April     | 0.60            | 5.89                                           |
| May       | 0.67            | 5.96                                           |
| June      | 0.71            | 5.96                                           |
| July      | 0.72            | 6.14                                           |
| August    | 0.71            | 6.57                                           |
| September | 0.69            | 6.94                                           |
| October   | 0.57            | 6.95                                           |
| November  | 0.51            | 6.21                                           |
| December  | 0.61            | 6.06                                           |

energy system from May to September [52]. The period ranging from June to September is now generally considered the dry season. The minimum temperature is experienced in July at around 6.10 °C, while the maximum temperature is in

November at about 26.80 °C [52]. Table 9 indicates collected data for meteorological information (2000–2017) [52]. Table 8 and Figure 7 indicate data for rainfall and temperature for the proposed site of study.

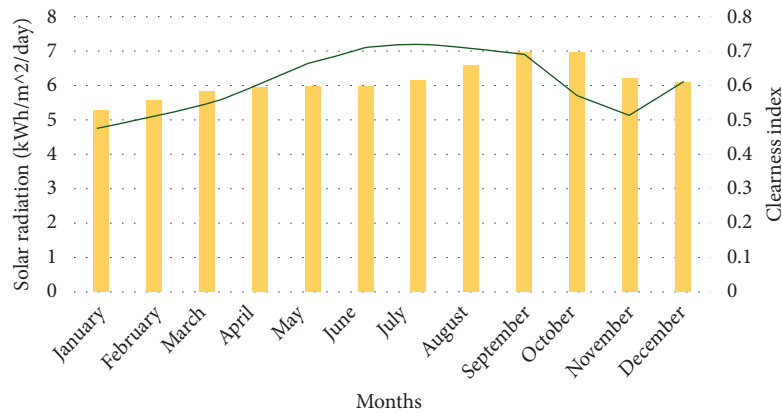


FIGURE 6: Potential of solar energy for the proposed site.

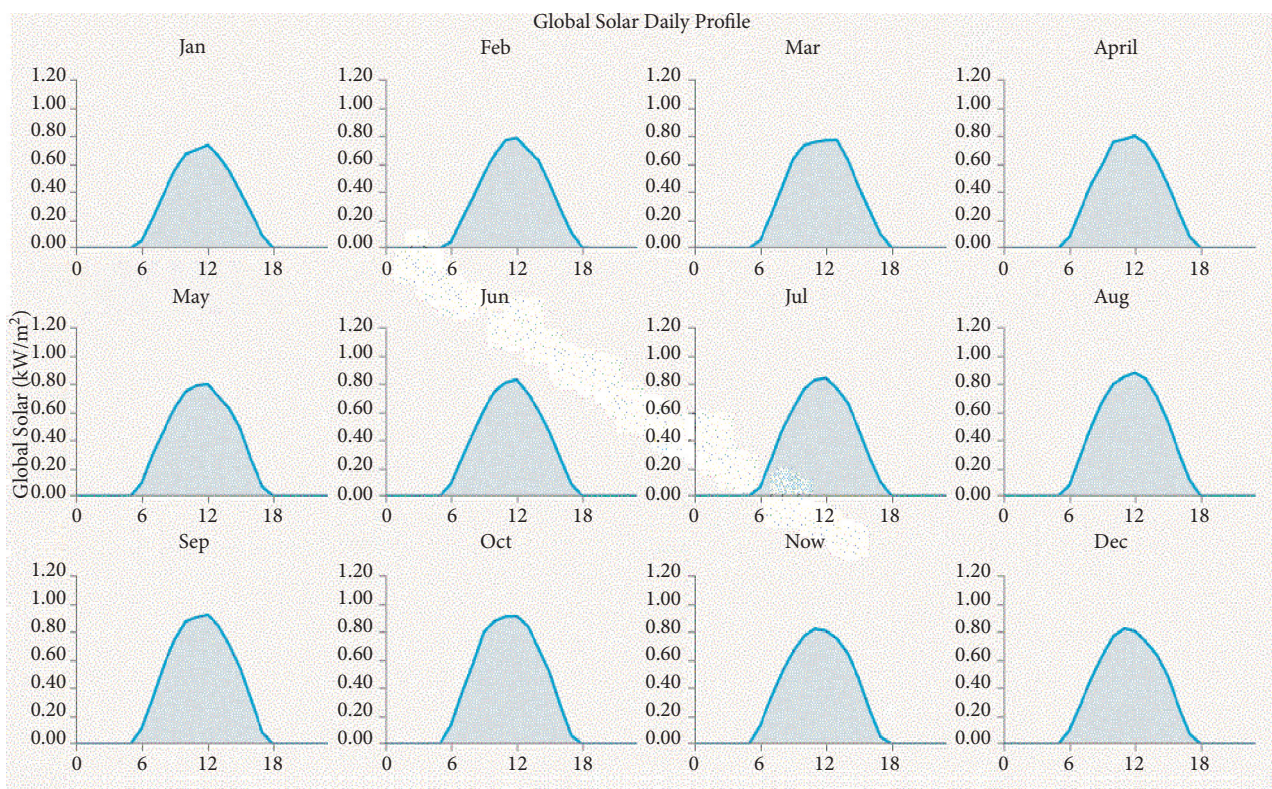


FIGURE 7: Annual global solar energy resource.

**4.4. Electrical Load Demand.** Ikukwa Village, Mbeya Rural District, Mbeya Region of Tanzania, consists of 14 small communities. The load demand in this area of study is found in four categories such as residential, institutional, small commercial, and institutional and small-scale industrial loads. Normally, in most of the rural and remote areas electrical load demand is dominated by the residential load. As shown in Table 2, only 7% of all communities have access to grid connectivity, particularly domestic houses. This limited grid coverage is due to scattered population and shortage of budget. This study considers only 7 communities without access to electricity. Other remaining small communities with at least access are not considered on the

assumption that all loads are connected to the distribution system of the grid. Therefore, data collection for the electrical energy demand of small communities without electricity for 100% is presented in Table 10. Electrical demand for such communities has been estimated by collecting data on rated power of anticipated appliances. In the first place, these data have been deliberately distributed over twenty-four-hour period on the excel sheet. The maximum estimated daily load profile is around 30 kW. Moreover, similar data for the communities with limited access to electricity have been implemented in HOMER Pro software. Electrical load profiles in tropical areas such as Tanzania are considered the same every month of the year. The reason why is

TABLE 9: Collected data for meteorological information (2000–2017) [52].

| Month     | Avg. Rainfall in (mm) | Max. temperature in °C | Min. temperature in °C |
|-----------|-----------------------|------------------------|------------------------|
| January   | 247.78                | 24.10                  | 14.70                  |
| February  | 207.78                | 24.40                  | 14.50                  |
| March     | 150.50                | 24.40                  | 14.30                  |
| April     | 117.64                | 23.50                  | 12.80                  |
| May       | 11.60                 | 23.10                  | 10.00                  |
| June      | 0.00                  | 22.30                  | 7.10                   |
| July      | 0.00                  | 22.40                  | 6.10                   |
| August    | 0.00                  | 24.00                  | 7.70                   |
| September | 0.00                  | 26.00                  | 10.80                  |
| October   | 16.70                 | 26.00                  | 12.30                  |
| November  | 78.10                 | 26.80                  | 14.50                  |
| December  | 175.20                | 24.90                  | 24.90                  |

TABLE 10: Total electrical load for small communities at Ikukwa Village without access to electricity (source: survey by authors before 31.07.2019).

| Name of the community without electricity | Estimated load in watts (W) |
|-------------------------------------------|-----------------------------|
| Juakali                                   | 98072                       |
| Kiwanja                                   | 52306                       |
| Kariakoo                                  | 79475                       |
| Mbuwi                                     | 80626                       |
| Mdonya                                    | 70790                       |
| Ukwaheri                                  | 46420                       |
| Unguja                                    | 33 404                      |
| Total                                     | 461093                      |

that there is no extreme weather variation. The daily random variability has been assumed to be zero. The estimated Ac load (baseline) for the site is as follows: average energy is 511.1 kWh/day, average power capacity is 21.3 kW, peak power is 30.31 kW, and load factor is 0.71. Figures 8–10 indicate the daily electrical load profile for the sub-villages without electricity connectivity from the national grid obtained by excel datasheet, daily electrical load profile simulated by HOMER Pro software, and monthly average seasonal load profile, respectively.

In the first place of this study, the analysis for the proposed system comprises the traditional approach of measurements and soft computing technique (HOMER software). In the absence of measured and financial limitations, traditional methods of measurements have been used to determine the profile for the tributary under study. This traditional approach of measurements has been used to determine the water flow rate and head, which are important parameters for an approximation of expected maximum generated output from the MHPS resulting during the rainy season.

MHPS is amalgamated with solar PV and batteries constituting the proposed hybridized energy system. Then, the traditional approach has been validated in HOMER software to acquire the optimized configuration of the proposed system at the lowest total NPC. This type of software is mostly used for optimization and sizing of the hybrid energy system. Nevertheless, optimization by the aforesaid software has some limitations such as limited

multiple objective functions; results of the COE are not arranged correspondingly in alignment with the results of total NPC, no consideration of depth of discharge (DOD), hourly variations of the parameters, and fluctuations of bus electrical energy profile. To deal with the abovementioned drawbacks of using HOMER software for optimization purpose, researchers have been constantly modernizing optimization methods. These advanced optimization techniques are termed as meta-heuristic methods. The grey wolf optimization (GWO) technique is one of the meta-heuristic methods. The merits of GWO approach are simplicity and adaptability [49, 56].

Therefore, in addition to soft computing optimization using HOMER software, the optimization of the total NPC has been carried out using AI techniques such as PSO, GWO, and GWO-PSOHD algorithms for obtaining supplementary optimal results. These AI methods are also used to minimize the total NPC of the proposed hybrid power technology according to equation 29. The optimization of the system using the AI approaches has been executed in MATLAB with specified input parameters.

## 5. PSO Algorithm

The PSO algorithm is one of the stimulated stochastic population-based optimization approaches established since 1995. The developers of this algorithm are Kennedy and Eberhardt, and its application reflects behavioural aspects of birds' flock. Basically, the PSO application involves the simulation of birds' social behaviour of assembling and adjusting themselves in their location while looking for attractive foodstuffs, defending against other greedy beasts using the reciprocated information and communal sensitive intelligence. The benefits of the PSO approach against other investigative optimization techniques are uncomplicatedness, straightforwardness regarding usage, great merging level, low storage needs, and reduced reliance on the preliminary points [57].

The PSO algorithm consists of both individual familiarity and acquired knowledge from working in the gathering to develop the simplified model that can successfully give solution to the complicated problems. This approach employs several agents (also known as particles), which build up the group going around exploring for the finest



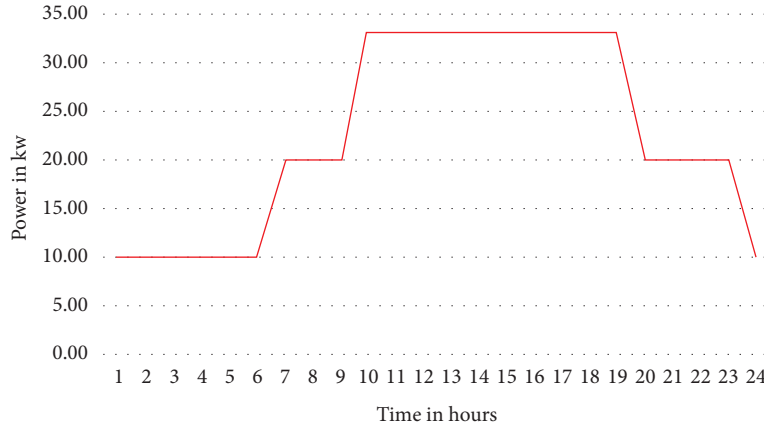


FIGURE 8: Electrical load profile for small communities without electricity connectivity from the national grid at Ikukwa Village, Tanzania.

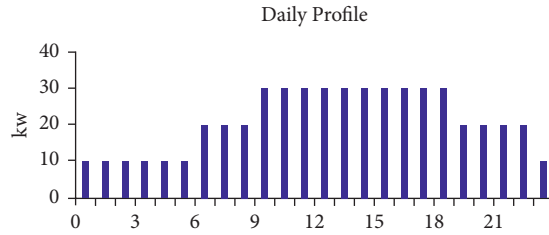


FIGURE 9: Electrical load over a 24-hour period simulated by HOMER Pro software.

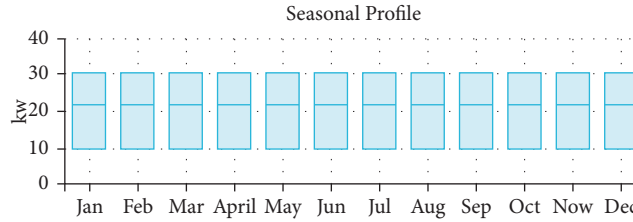


FIGURE 10: Monthly average seasonal load profile.

answer. Every agent in the exploring cosmos alters its hovering location based on its individual flying knowledge plus the flying practice of other agents in the crowd. Every agent maintains its distinct position in the place of solution. The coordinates are related to its finest fitness rate. This rate is known as personal best ( $P_{best}$ ). In addition to the  $P_{best}$ , there is another finest rate for fitness, which is also taken into account as the best fitness rate in the PSO algorithm, and it is acquired by any agent in proximity to that individual agent. This kind of the finest fitness rate is named as global best ( $G_{best}$ ) [57].

**5.1. Mathematical Representation of PSO Algorithm.** In this part, the mathematical model of the PSO approach is defined. The primary notion of the PSO approach is built on the speeding every particle in the direction of its  $P_{best}$  plus  $G_{best}$  positions at a randomly weighted fast-moving factor at every period of time. Each particle tends to modify its

coordinates according to some equation and variables. The present location, speeds, and the distance of the  $P_{best}$ , and the distance between the available location of  $G_{best}$  are modified. The variation of speed and location of the particle can be represented numerically as follows [57]:

$$\begin{pmatrix} V_{k+1}^i = V_k^i + C_1 r_1 (p_k^i - x_k^i) + C_2 C_2 (p_k^i - X_k^i) \\ X_{k+1}^i = X_k^i + V_{k+1}^i \end{pmatrix}, \quad (13)$$

where  $V_k^i$  symbolizes the position of the particle,  $V_k^i$  delineates the velocity of the particle,  $p_k^i$  stands for the finest recalled location,  $C_1$  and  $C_2$  describe the reasoning and shared variables, and  $r_1$  and  $r_2$  are randomized numbers ranging from 0 to 1.

**5.2. The Primary PSO Algorithm.** The key steps in the PSO approach are provided as follows [58]:

- (i) Initializing the population size of the particles
- (ii) Reiterating
- (iii) Computing the function's fitness for personal particle
- (iv) Adjusting the best particle in the swarm
- (v) Picking out the finest particle
- (vi) Modernizing the particle's location

Reiterating up to when the required conditions are fulfilled.

Apart from other AI optimization methods, in this research paper the PSO application has been employed to assess the monetary metrics such as total NPC and its parallel COE based on (6) and (7). The PSO algorithm has been executed with a fixed mass of inertia, and input parameters are described in Table 11. The location of a particle inside the search space embodies the power capacities of apparatuses of a given structure of energy system. For a single generation, this PSO method regulates the location of the particles in the swarm, tending to move in the direction of the optimum solution. Every particle in the swarm has the capability of understanding its existing location and speed, that is, its own finest location and the finest location of whole group for a single generation. Deployment of the input technical specifications and power capacity equivalent to the location of an individual particle is estimated for a period of time of one year. Then, in case all specified limits are met, either the fitness of energy system or the total NPC is computed using input economic metrics. Violation regarding any limit by energy system configuration is instantly rejected. Practically, NPC is adjusted to the huge quantity so as to limit the group from being pulled to this location inside the exploration space. The outcomes of the preceding generation are inputted to the succeeding generation. The procedure of the PSO method is repetitive until the greatest quantity of generations is attained. Finally, the solution for NPC is acquired at optimal power capacity [58].

## 6. GWO Algorithm

In the second phase of this study, further optimization results for the proposed energy system are obtained from the GWO technique. The GWO procedure for the first time was suggested by Mirjalili and other researchers in 2014. The key motivation of this approach is the management and pursuing technique of grey wolves. In the mathematical expression, the first group that is known as alpha ( $\alpha$ ) wolves has been reflected as the best fitting solution. Consequently, beta ( $\beta$ ) and delta ( $\delta$ ) are the second and third finest solutions, respectively. The remaining fourth candidate solution is known as omega ( $\omega$ ) wolves.  $\alpha$ ,  $\beta$ , and  $\delta$  wolves lead to hunting in the GWO optimization, while the group of  $\omega$  wolves follows the first three packs during searching for the optimum global solution. The societal pyramid of all groups of wolves is illustrated in Figure 11.

Moreover, the following equations are presented for enclosing and chasing the wolves [49, 56, 58]:

TABLE 11: Input parameters of PSO approach.

| Input variables          | Worth  |
|--------------------------|--------|
| Least search space       | 0      |
| Greatest search space    | 10 000 |
| Quantity of iterations   | 100    |
| Size of population       | 50     |
| Quantity of generations  | 200    |
| Weightiness of inertia   | 0.6    |
| Weightiness of cognition | 1.6    |
| Weightiness of community | 1.6    |

$$\vec{D} = |\vec{C} \cdot \vec{Xp}(t) - \vec{X}(t)|, \quad (14)$$

$$\vec{X}(t+1) = \vec{Xp}(t) - \vec{A} \cdot \vec{D}. \quad (15)$$

In these given equations,  $\vec{Xp}$ ,  $\vec{X}$ , and  $t$  define the position vectors of the prey, grey wolf, and current iteration, respectively. Then, coefficient vectors are denoted by  $\vec{A}$  and  $\vec{C}$ , and they are estimated using the following equations [49, 56, 58]:

$$\vec{A} = 2\vec{a}\vec{R1} - \vec{a}, \quad (16)$$

$$\vec{D} = 2\vec{R2}. \quad (17)$$

In the above equations, uneven vectors are denoted as  $\vec{R1}$  and  $\vec{R2}$ . They exist in the range starting from 0 to 1, while  $\vec{a}$  is diminished from 2 to 0 in the complete iteration.

The wolves may identify the locality of the quarry and surround them directed by  $\alpha$ . Yet, the location of the optimum quarry is not recognized in exploration space. Mathematically, the model for pursuing the performance of the wolves is marked as  $\alpha$ ,  $\beta$ , and  $\delta$ . They signify the grey wolves knowing fully the potential whereabouts of the prey. Therefore, the foremost three finest answers are stored. Then,  $\omega$  group of the wolves needs to make the revision of their sites more improved than the present ones as defined in the given equations [49, 56, 58]:

$$\vec{D\alpha} = |\vec{C1} \cdot \vec{X\alpha} - \vec{X}|,$$

$$\vec{D\beta} = |\vec{C2} \cdot \vec{X\beta} - \vec{X}|,$$

$$\vec{D\delta} = |\vec{C3} \cdot \vec{X\delta} - \vec{X}|,$$

$$\vec{X1} = \vec{X\alpha} - \vec{A1} \cdot (\vec{D\alpha}), \quad (18)$$

$$\vec{X2} = \vec{X\beta} - \vec{A2} \cdot (\vec{D\beta}),$$

$$\vec{X3} = \vec{X\delta} - (\vec{A3} \cdot (\vec{D\delta})).$$

$$\vec{X}(t+1) = \frac{(\vec{X1} + \vec{X2} + \vec{X3})}{3}.$$

When the quarry halts from moving, the process of hunting the wolves also terminates. Referring to the given

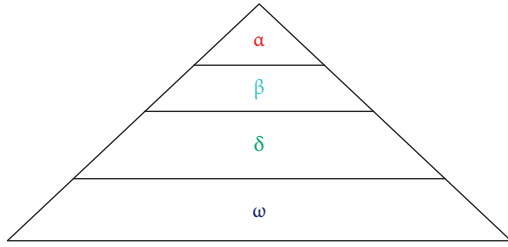


FIGURE 11: Social ladder of grey wolves.

equations,  $\vec{a}$  diminishes from 2 to 0 and therefore  $A$  also decreases since  $\vec{A}$  is dependent of  $\vec{a}$ . The quarry is attacked by grey wolves if  $\vec{A}$  exists in the range starting from  $-1$  to  $1$ . Moreover, weights for the definition of distance by the quarry are varied by  $\vec{C}$ . This value is important for saving answer from the local bests, specifically in the final iterations. General steps executed in the GWO algorithm are summarized as follows [49, 56]:

- (i) The establishment of the population of grey wolf  $X_i$  ( $i = 1, 2, 3, \dots, n$ )
- (ii) The initialization of  $a$ ,  $A$ , and  $C$
- (iii) The computation of the fitness of each pursuit agent, that is,  $X_\alpha$ ,  $X_\beta$ , and  $X_\delta$ .  $X_\alpha$ ,  $X_\beta$ , and  $X_\delta$  stand for the finest pursuit agent, the 2nd finest pursuit agent, and 3rd finest pursuit agent, respectively
- (iv) While ( $t < \text{Maximum quantity of iterations}$ )
- (v) For every pursuit agent
- (vi) The modernization of the location of the present pursuit agent
- (viii) End for
- (ix) The modernization of the  $\alpha$ ,  $A$ , and  $C$
- (x) The computation of the fitness of all pursuit agents
- (xi) Modernization of the  $X_\alpha$ ,  $X_\beta$ , and  $X_\delta$
- (xii)  $T = t + 1$
- (xii) End while
- (xiii) Return  $X_\alpha$

**6.1. GWO-PSOHD Algorithm.** The GWO-PSOHD approach of optimization is a recently innovated nature-inspired technique. It is one of the hybridized categories of the GWO. GWO is deliberately combined with the PSO method for the refinement of the working capability of the GWO. It is not a scope of this specific study to outline, compare, and contrast the applications of all types of the hybridized forms of GWO. Nevertheless, the hybrid-based GWO algorithm has indicated superior results, specifically convergence level, improved outcomes, effectiveness, and greatest steadiness in more discrete algorithms for  $n$ . It has the ability of handling both controlled and uncontrolled optimization problems. This method focuses on the optimum solution by holding the improved solutions and terminates the furthestmost ones [59, 60].

In this study, a new method called GWO-PSOHD is investigated for the optimization of the total NPC and corresponding leveled COE. Therefore, this study concentrates on the optimization by determining the least total NPC of renewable-based micro-grid of multiple sources.

**6.2. Fundamental Practices of GWO-PSOHD Approach.** The following are the steps for the GWO-PSOHD approach [59, 60]:

- (i) Initializing the search agents and describing the solution space
- (ii) Running the GWO method
- (iii) Generating the least magnitudes for the complete number of agents
- (iv) Permitting the agents to the PSO method
- (v) Returning the adapted location back to the GWO method
- (vi) Repeating similar procedures up to when the ending conditions are acquired.

## 7. Modelling of Individual Components Incorporated in the Proposed Hybrid Micro-Grid System

If the total generated power by energy sources integrated into the hybrid micro-grid system equals energy requirement, this means the total generated power meets the electrical load. In case, the generated electricity is greater than the load demand, and surplus electricity is stowed in the batteries (battery charging process). If the energy requirement is greater than the generated electricity, the stored energy is derived from the batteries (discharging process) to meet the energy requirement. In some conditions, energy demand is greater than the sum of generated electricity and stored energy. In such a situation, the portion of the load demand needs to undergo shedding, thus resulting in a loss of electricity supply. The proposed hybrid system incorporates main components such as MHT, solar panel, batteries, converter, and electrical load profile. Based on these components plus electric cables, MHPS/solar PV/battery hybrid micro-grid system is formed. Therefore, the technical and economic investigation may be carried out. According to the power generation and electrical load profile, reliability and economic functions can be formulated. In other words, power generated from the system in the off-grid mode is required to reliably meet the anticipated electrical load profile at the lowest cost. This power generated from renewable sources is expressed as follows [49]:

$$P_{\text{ghybrid}} = P_{\text{MHPS}} + P_{\text{SolarPV}}, \quad (19)$$

where  $P_{\text{ghybrid}}$  is the total power generated from the hybrid micro-grid system (W),  $P_{\text{MHT}}$  is the power generated by one turbine (W), and  $P_{\text{solar PV}}$  is the power output from solar PV (W). The power  $P_{\text{ghybrid}}$  produced by the renewable energy sources is required to generate sufficient power for the energy requirement plus losses. Energy storage, especially



batteries, is required to counteract the intermittent nature of renewable energy sources.

The power generation and overall performance of the entire hybrid system are an outcome of the operation of discrete component comprised in the system. Thus, the modelling of the components amalgamated in the system is as follows [54, 61].

**7.1. Modelling of MHT.** The potential of hydropower depends on the two primary input parameters such as water flow rate and water head. Generated power output from a single water turbine is computed using the following formula [40, 61]:

$$P_{MTH} = \frac{\eta_{MHT} \times \rho \times g \times H \times Q_{MHT}}{1000}, \quad (20)$$

where  $P_{MHT}$  is the power produced by a single turbine (kW),  $\eta_{MHT}$  is the water turbine efficiency (%),  $\rho$  is the density of water ( $\text{kg/m}^3$ ),  $H$  is the water head (m), and  $Q_{MHT}$  is the water flow rate flowing into the water turbine ( $\text{m}^3/\text{s}$ ).

As a matter of fact, hydropower turbine has practical and flow limits. It can operate effectively within certain flow ranges. If only one turbine is used in a hydropower system, three different operation stages can be distinguished as follows [40, 61]:

$$Q(t) = \begin{cases} 0, & \text{if } Q(t) \leq Q_{Mn} \\ Q, & \text{if } Q_{Mn} \leq Q(t) \leq Q_{Mx} \\ Q_{Mx}, & \text{if } Q_{Mx} \leq Q(t) \end{cases} \quad (21)$$

where  $Q_{Mn}$  is the minimum flow rate, cutoff, below which electricity generation stops, and  $Q_{Mx}$  is the maximum flow rate a turbine can generate electricity, and  $Q(t)$  is the existing water flow rate.

The total power produced by the MHPS is calculated as follows [40]:

$$P_{MHPS} = N_{MHT} \times P_{MHT}, \quad (22)$$

where  $P_{MHPS}$  is the total power generated by the microhydropower system (ROR hydropower generation systems without pondage),  $N_{MHT}$  is the total number of water turbines, and  $P_{MHT}$  is the approximated power by one hydro-

water turbine. Table 12 shows the summarized technical and fiscal specifications for MHT.

**7.2. Modelling of Solar Photovoltaic (Solar PV).** The solar PV system is designed by computing the suitable amount of solar panels. These solar panels transform solar radiation into direct current electricity. The solar panel is an arrangement of PV cells. Generated output per panel  $P_{SPV}$  at a given time in hours is defined as follows [54]:

$$P_{SPV}(t) = SOI(t) \times A_{SPV} \times \eta_{SPV}, \quad (23)$$

where  $SOI$  refers to the solar insolation in  $\text{kW/m}^2$  at a given time  $t$  in hours,  $A_{SPV}$  is the surface area of a single solar panel ( $\text{m}^2$ ), and  $\eta_{SPV}$  signifies the efficiency of solar panel (%). In this research work, solar panels are assumed to consist of the maximum power point tracking scheme and the effects of ambient temperature are negligible.

The total power output generated by a solar panel is estimated as follows [54]:

$$P_{\text{solar PV}}(t) = N_{SPV} \times P_{SPV}, \quad (24)$$

where  $P_{\text{solar PV}}$  is the total power generated by solar panels in the proposed hybrid system. In this research work, the CSUN 410–144M Model of solar panel with peak power capacity of 410 W has been used in the performance analysis of the hybrid system. Table 13 illustrates the summarized technical and fiscal specifications for adopted solar PV.

**7.3. Modelling of the Battery.** Deep cycle batteries are mostly favoured for the storage of electricity for off-grid applications, particularly lead-acid batteries [54]. A detailed analysis of charging and discharging conditions of batteries is vital for the determination of finest capacity. The SOC ( $t$ ) of a set of batteries at a time of one hour is related to SOC ( $t-1$ ), the preceding condition of the charge, and the energy exhaustion or accrual state from  $t-1$  to  $t$  [54].

If  $P_{\text{hybrid}}(t) = P_{\text{solar PV}}(t) + P_{MHT}(t) \geq PL(t)/\eta_{INV}$ , generated electricity from given renewable energy sources is sufficient for meeting load demand  $PL(t)$  plus extra energy kept in the batteries. At this state, the batteries are being charged. The SOC ( $t$ ) is defined in the following expression [54]:

$$SOC(t) = SOC(t-1) \times (1 - \sigma) + \frac{P_{\text{hybrid}}(t) - PL(t)/\eta_{INV} \times \eta_{bc}}{1000 \times Nb \times Cb}, \quad (25)$$

where  $\eta_{INV}$  refers to the inverter efficiency,  $\eta_{cb}$  is the charging efficiency of the batteries,  $C_b$  is the nominal size of the battery in kWh,  $N_b$  is the quantity of batteries, and  $\sigma$  is self-discharging rate per hour.

Similarly, in case  $P_{\text{hybrid}}(t) < (PL(t))/(\eta_{INV})$ , generated electricity from renewable energy sources is not sufficient

and therefore does not satisfy electrical load demand. Energy stored in the batteries is discharged to balance the power supply and demand. For this condition, SOC ( $t$ ) is expressed as follows [54]:

$$SOC(t) = SOC(t-1) \times (1 - \sigma) - \frac{PL(t)/\eta_{INV} - P_{\text{hybrid}}(t)/\eta_{db}}{1000 \times Nb \times Cb}, \quad (26)$$

TABLE 12: Summarized technical and fiscal specifications for MHT.

| Parameter                                       | Specifics                |
|-------------------------------------------------|--------------------------|
| Type of electric generator                      | Alternating current (AC) |
| Generated power by MHT                          | 8–40 kW                  |
| Efficiency of the turbine                       | 75%                      |
| Estimated head of turbine                       | 13 m                     |
| Flow rate channel                               | 160 l/s                  |
| Minimum flow rate                               | 60%                      |
| Maximum flow rate                               | 115%                     |
| Life span of the project                        | 75 years                 |
| Charge for capital/unit                         | 50, 000 USD              |
| Charge for replacement/unit                     | 43, 000 USD              |
| Charge for operation and maintenance (O&M)/unit | 5, 000 USD/year          |

TABLE 13: Summarized technical and fiscal specifications for adopted solar PV.

| Parameter                                     | Specifics            |
|-----------------------------------------------|----------------------|
| Company                                       | CSUN                 |
| Panel model                                   | CSUN 410–144M        |
| Solar technology                              | Monocrystalline semi |
| Power capacity pmax                           | 410 W'               |
| Temperature range                             | −40°C–80°C           |
| Efficiency                                    | 20.31%               |
| Dimensions                                    | 1996 * 991 * 35 mm   |
| Estimated area of the panel                   | 1.98 m <sup>2</sup>  |
| Weight                                        | 22.9 Kg              |
| Tc of voltage                                 | −0.30%/° C           |
| Tc of current                                 | +0.06%/° C           |
| Tc of power                                   | −0.39%/° C           |
| Life span of the project                      | 25 years             |
| Charge for capital/kW                         | 2500 USD             |
| Charge for replacement/kW                     | 2500 USD             |
| Charge for operation and maintenance (O&M)/kW | 10 USD/year          |

Tc = temperature coefficient.

where  $\eta_{db}$  represents the discharging efficiency of batteries. Excessive charge ejection needs to be evaded in order to ensure a longer lifespan of batteries and SOC (t) should be subjected to the following restrictions [54]:

$$(1 - \text{Max.DOD}) \leq \text{SOC}(t) \leq 1, \quad (27)$$

where Max. DOD is the maximum depth of discharge. Alternatively, [40]

$$\text{SOC}_{\min} \leq \text{SOC}(t) \leq \text{SOC}_{\max}, \quad (28)$$

where  $\text{SOC}_{\min}$  and  $\text{SOC}_{\max}$  are lowest and highest value states of charge of the battery storage.

In this study, the Just Gel 12–100 Model of the battery is chosen for the performance analysis of hybrid system. Table 14 portrays technical and fiscal specifications for battery energy storage bank.

**7.4. Modelling of the Converter.** The converter is an electric device used for maintaining the flow of electricity in a renewable energy-based hybrid system. The converter comprises inverter and rectifier for conversion of DC power to

AC and vice versa, respectively. Power from the output of converter should be higher than the maximum load. In this study, the power output of MHPS is AC that is connected to AC bus and power output of solar PV and battery energy storage bank is DC. Therefore, the inverter is essential for meeting electrical load demand. The rated power for the device is defined as follows [54]:

$$PL = P_{\text{inverter}} = \frac{P_{\max AC}}{\eta_{\text{inverter}}}, \quad (29)$$

where  $P_{\text{inverter}}$  is the rated power of inverter,  $P_{\max AC}$  is the maximum AC electrical load, and  $\eta_{\text{inverter}}$  is the efficiency of the inverter. Table 15 portrays the technical and economic parameters for adopted converter.

## 8. Optimization Models of the Proposed System Using GWO Algorithm

The MHPS/solar PV/battery bank hybrid system is optimized by taking into account different parameters such as loss of power supply probability (LEPP) for the system

TABLE 14: Technical and fiscal specifications for battery energy storage bank.

| Parameter                                  | Specifics       |
|--------------------------------------------|-----------------|
| Company                                    | Zhejiang Just   |
| Type of the model                          | Just Gel 12–100 |
| Weight                                     | 30.5 Kg         |
| Operating temperature                      | 25°C ± 3°C      |
| Nominal voltage                            | 12 V            |
| Nominal capacity                           | 100 AH          |
| Peak capacity                              | 100 AH          |
| Efficiency of charging                     | 90% (assumed)   |
| Efficiency of discharging                  | 95% (assumed)   |
| Highest charging current                   | 20 A            |
| Highest discharging current                | 800 A           |
| Depth of discharge (DOD)                   | 80% (assumed)   |
| Rate of self-discharge per hour            | 3%              |
| Charge for capital                         | USD 200         |
| Charge for replacement                     | USD 200         |
| Charge for operation and maintenance (O&M) | USD 10          |
| Life span of the project                   | 15 years        |

TABLE 15: Technical and fiscal specifications for converter.

| Parameter                                       | Specifics  |
|-------------------------------------------------|------------|
| Rated capacity                                  | 40 KW      |
| Inverter efficiency                             | 90%        |
| Inverter efficiency                             | 85%        |
| Charge for capital/unit                         | 2000 USD   |
| Charge for replacement/unit                     | 2000 USD   |
| Charge for operation and maintenance (O&M)/unit | 0 USD/year |
| Life span of the project                        | 25 years   |

reliability, total NPC, and COE. The sizing optimization of the system is obtained by the minimization of the NPC and also COE using the GWO algorithm. The optimization problem involves the formulation of objective function and constraints. In this study, the optimization of the objective function is performed under the restrictions such as upper and lower boundaries based on the amount of specified components and their power capacities, battery storage restrictions, and SPS for the technical and economic analysis of the system. The objective function under subjected limitations is formulated as follows [40, 49, 54].

**8.1. Objective Function with Defined Constraints.** The main purpose of the optimization is to attain the bottommost cost of hybrid power-generating system using the GWO method. The objective function is formulated to minimize the total NPC as follows [40, 49, 54, 62]:

$$\text{Total NPC} = \sum_{i=(N_{\text{SoPV}}+P_{\text{MHPS}}+N_{\text{BTR}}+N_{\text{CONV}})} \frac{TANC_i}{CRP}. \quad (30)$$

The minimization of the total NPC according to the objective function is executed under the following constraints as follows:

$$\begin{aligned} N_{\text{SOPV}}^{\text{mini}} &\leq N_{\text{SOPV}} \leq N_{\text{SOPV}}^{\text{maxi}}, \\ P_{\text{MHT}}^{\text{mini}} &\leq P_{\text{MHT}} \leq P_{\text{MHT}}^{\text{maxi}}, \\ N_{\text{BTR}}^{\text{mini}} &\leq N_{\text{BTR}} \leq N_{\text{BTR}}^{\text{maxi}}, \\ N_{\text{CONV}}^{\text{mini}} &\leq N_{\text{CONV}} \leq N_{\text{CONV}}^{\text{maxi}}, \\ LEPP^{\text{mini}} &\leq LEPP \leq LEPP^{\text{maxi}}, \end{aligned} \quad (31)$$

where  $N_{\text{SOPV}}$ ,  $N_{\text{BTR}}$ , and  $N_{\text{CONV}}$  are quantities of solar panels, batteries, and converters, respectively  $P_{\text{MHT}}$  is the power produced by micro-hydropower units forming  $P_{\text{MHPS}}$ .  $P_{\text{MHPS}}$  is power generated by MHPS from the tributary under study. Superscript maxi and subscript mini indicate maximum and minimum values of given parameters.

**8.2. The Highest and Lowest Power Capacities of MHT.** The water resource of Ikata tributary has been comprehensively discussed in Section 3. The model for MHPS has been executed in the HOMER Pro software. It should also be remembered that the range of the overall flow rate of the Ikata tributary starts from 160 l/s to 2.943 m<sup>3</sup>/s. The minimum flow rate results during the dry season; thus, no power is produced from the tributary. So long, there is sufficient water flow rate during the rainy season and head, and the designed MHPS is capable of meeting the total actual load of 40 kW (greater than daily peak load). In this study, the turbine with an inconstant speed has been taken into consideration to maximize the use of the micro-hydropower potential within the given entire flow rate range and water head. The power capacity of the MHT can be estimated according to equation (23). Input parameters for sizing the MHT are tabulated in summary Table 12. In this speed adjustability of MHT, the minimum power capacity ( $P_{\text{MHT}}^{\text{mini}}$ ) at time in hours is anticipated to be twenty percent of the highest turbine generation capacity ( $P_{\text{MHT}}^{\text{maxi}}$ ), which balances the estimated actual load. The range of variable power from the MHT is expressed as follows [40, 54]:

$$8 \text{ kW} \leq P_{\text{MHT}}(t) \leq 40 \text{ kW}. \quad (32)$$

**8.3. The Highest and Lowest Number of Components.** The number of the solar photovoltaic panels, batteries, and converters may change to fulfil power supply requirement. The highest and lowest boundaries of the aforesaid components are presented in equations as follows [40, 63].

The maximum number of solar panels is computed as follows: in the first place, data for solar irradiance (1 kW/m<sup>2</sup>), area (m<sup>2</sup>), and efficiency of the model of the solar panel (20.31%) are tabularized in Table 13 to estimate power output produced by a single panel using equation (41). Therefore, [63]

$$P_{\text{SPV}} = 0.402 \text{ kW}. \quad (33)$$

The maximum number of panels for the generation power capacity balancing the expected actual load is approximated as follows [63]:



$$N_{SOPV}^{maxi} = \frac{PL}{P_{SPV}} = \frac{40 \text{ kW}}{0.402} \approx 100. \quad (34)$$

The maximum number of batteries can be estimated using the following equation. In the very beginning battery capacity,  $C_{BTR}$  is determined [63, 64] as follows:

$$\begin{aligned} C_{BTR} &= \frac{\text{Daily load (Wh)} \times D_D}{\text{System voltage } V \times \eta_{BTR} \times \max. \text{ DOD}} \\ &= \frac{511100 \times 7}{48 \text{ V}} \text{ AH} \\ &= \frac{74, 527 \text{ AH}}{48 \text{ V}} \\ &= 10, 647 \text{ AH}. \end{aligned} \quad (35)$$

The autonomy ( $D_D$ ) of 7 days gives the total battery capacity equals 74, 529 AH. Max. DOD and overall battery efficiency are both assumed to be 100% So, the highest number of batteries can be computed using the following equation [63, 64]:

$$N_{BTR}^{maxi} = \frac{\text{Total esstimated capacity}}{\text{Nominal battery capacity}} = \frac{74, 529 \text{ AH}}{100 \text{ AH}} \approx 746. \quad (36)$$

The protection against the excessive charge and over-discharge of the battery should be done for securing it from damage. The highest and lowest bounds of system storage may also be considered as boundaries. The boundaries are also directly proportional to the available number of batteries and SOC. The limits for battery bank capacities are given as follows [62]:

$$e_{BMI} \leq e_B \leq e_{BMA}, \quad (37)$$

where  $e_{BMI}$  and  $e_{BMA}$  are the least and greatest magnitudes of capacities of battery bank. These storage capacities are directly proportional to the available number of batteries and SOC and are computed in the following equations [62]:

$$e_{BMI} = \frac{N_{BTR} \times V_N \times C_{BTR}}{1000} \times SOC_{mini}, \quad (38)$$

$$e_{BMA} = \frac{N_{BTR} \times V_N \times C_{BTR}}{1000} \times soc_{maxi}, \quad (39)$$

where  $V_N$  is the nominal voltage of the battery (V),  $C_{BTR}$  is the battery size (AH), and estimated values for  $soc_{mini}$  and  $SOC_{maxi}$  are 0.2 and 1, respectively.

The highest number of converters is the calculated assumption that a single converter has rated capacity of about 2000 kW.

Thus, the highest number of converters is estimated as follows:

$$N_{CONV}^{maxi} = \frac{\text{Total actual load}}{\text{Power capacity of one converter}} = \frac{40 \text{ kW}}{2000} = 20. \quad (40)$$

**8.4. The Highest and Lowest of LEPP.** LEPP can be computed using (34) after the determination of the shortage of power supply using equation (35). For the straightforwardness of calculating the LEPP, in this study the equation (4) can be simplified into the following equation [49, 65]:

$$SPS = P_L - (P_{ghybrid} + P_{battery \text{ bank}}). \quad (41)$$

If the power generated from renewable energy sources plus stored power in the battery bank satisfies all energy required for 100%, (4) rewritten as (42). LEPP can be estimated using equation (3) as follows [40, 49, 54, 65]:

$$SPS = P_L - 100\% P_L. \quad (42)$$

$$SPS = 0. \quad (43)$$

LEPP is determined by substituting (43) into (3), which equals zero. It implies that the system satisfies the power supply requirement when LEPP = 0.

Using the GWO method, the appraisal of the objective function according to the delineated constraints is made. Sizes of components are selected within the stated limits. The identification of viable patterns of the hybrid system is done in relation to the preset magnitudes of LEPP. The best possible solutions are repeated in the GWO method. The structure is updated to estimate the minimum total NPC.

The input parameters of PSO approach are shown in Table 11.

**8.5. Methodology for Sizing the Proposed Hybrid Micro-Grid System Using the GWO Algorithm.** Optimization parameters such as solar radiation, micro-hydro-energy source, and ambient temperature, electrical load data, and assumed constants (datasheet) were used as input variables. Decision variables such as  $A_{pv}$  and  $A_{MHT}$  were estimated by the GWO algorithm for individual iteration. Then, the power from solar PV ( $P_{pv}$ ) and micro-hydro-power systems ( $P_{MHT}$ ) was estimated with variable inputs, assumed constant inputs, and approximated decision variables. To meet the energy requirements, power is considered to be generated by MHT (MHPS) and solar PV. However, if generated electricity from aforesaid energy sources does not satisfy the load demand, stored energy from batteries is used to balance power supply and demand. The standards of LEPP were calculated for the total time in hours for a year equivalent to 8760 hours. In case any value of LEPP has been greater than the prescribed one, the decision values were then recalculated. If there was no deviation from the predetermined value, an optimization procedure was implemented. Then, the optimal size of the hybrid system is obtained corresponding to the lowermost total NPC (also COE), which is computed at LEPP equals zero when no shortage of power supply. Table 16 indicates miscellaneous specifications for the GWO technique.

TABLE 16: Miscellaneous specifications for the GWO technique.

| Parameters                                                 | Value     |
|------------------------------------------------------------|-----------|
| Population size of grey wolves $n$                         | 12        |
| Maximum number of iterations, $iter_{max}$                 | 100       |
| Penalty factor, PP                                         | $10^{10}$ |
| Dimension, $d$                                             | 2         |
| Inflation rate                                             | 5%        |
| Interest rate                                              | 3.04%     |
| Life span of the project                                   | 25 years  |
| Highest capacity of micro-hydro-turbines $P_{MHT}^{max,i}$ | 40 kW     |
| Highest quantity of solar panels $N_{SPV}^{max,i}$         | 100       |
| Highest quantity of batteries $N_{BTR}^{max,i}$            | 764       |
| Highest quantity of converters                             | 20        |
| Highest LEPP, $LEPP^{max,i}$                               | 0.07      |

## 9. Results and Discussions for the Technical and Economic Analysis of Proposed Hybrid Renewable Energy-Based System

The optimal hybrid configuration indicates micro-hydro-energy, solar energy, energy storage system, and power conditioning devices. MHPS consists of main components such as water turbine and generator. SPVS comprises solar panels, charge controllers, and batteries and converter. HOMER has been used to optimize the hybrid energy system. Figures 12 and 13 indicate optimal configuration for the proposed off-grid hybrid energy-based system and schematic depiction for off-grid hybrid micro-hydro- and solar system, respectively.

The optimal hybrid energy system has been simulated using HOMER Pro software according to the technical and economic details. Simulation results are of two kinds, that is, technical and economic results. In this simulation load, the following has been used as a dispatch strategy. Therefore, the simulated results of the optimized renewable energy system are presented as follows.

In this section, the results of electrical power analysis that include solar and micro-hydro-resource outputs, electrical power output, electricity generation, and performance of individual systems/components within the proposed hybrid energy system have been presented. The system comprises MHPS, solar photovoltaic panels, batteries, and converter. The proposed hybrid system will specifically use Natel-free jet (MHT), generic solar photovoltaic, and PowerSafe SBS 1500 (batteries). The designed system has the capacity of supplying electricity to the proposed site of peak load of 30.3 kW for a period of 25 years and interest rate of 3.04%. The installed generation capacity of the designed system should be equal to or greater than the daily peak power in order to meet the load demand. To account for losses by ensuring reliability and availability of power, the designed system should have greater capacity than the daily peak power. Therefore, the installed generation capacity is approximated to be 48 kW. Technical specifications of the system architecture are given in Table 17. The proposed hybrid energy system generated excess electricity, unmet electrical load, and capacity shortage for the proposed hybrid energy system which are 128 483 kWh/year,

0 kWh/year, and 0 kWh/year, respectively. Table 18 indicates the generation, consumption, and renewable energy utilization. Miscellaneous technical and economic details for all components in the hybrid energy system are presented in Tables 19 and 20, respectively.

Additional related information or parameters for MHPS are that the lifespan for MHPS is assumed to be 75 years. The estimated theoretical generation power capacity of the MHPS is 375 kW considering the water head of 13 metres and flow rate of 2943 litres per second. The minimum flow rate is 160 litres per second. The maximum and minimum values of water flow of the tributary are depicted in Figure 4. Generally, the maximum power-generated output of MHPS occurs during the rainy season, particularly in January. However, practically, the power output drops down depending on the rainfall-dependent water flow profile of the tributary under study and other technological aspects. Technological factors are water discharge rates and the efficiency of the turbine. The maximum and minimum discharge rates are 115 and 60 percentages, respectively. The efficiency of MHT in this study is considered to be 75%. During the dry season, that is, in the absence or shortage of water flow in the tributary, MHPS does not produce any power output. In the period starting from June to September, no power is generated from the MHPS. Figures 14 and 15 show the annual daily profile of micro-hydropower output and annual monthly averages of hydropower output, respectively.

Specific technical and economic details are also presented in Tables 19 and 20. As indicated in Figure 6, the area of proposed site has sufficient solar energy potential. This potential alongside battery energy storage assures the availability and reliability of power supply. The lifespan of SPVS is approximated to be 25 years, and the system is ground-mounted with a derating factor of 90%. It is capable of providing power output throughout both rainy and dry seasons for the proposed hybrid energy systems. Figure 15 displays the annual daily profile of solar photovoltaic power output (Figure 16).

Similarly, the additional technical and fiscal details for battery energy system are also tabulated in Tables 19 and 20. The life span of battery has been assumed to be 15 years; the battery energy system employs a total of 288 batteries of PowerSafe SBS 1500. The nominal voltage for the battery is 12 volts, and bus voltage is 48 volts (there are 4 batteries in series/per string and 24 parallel strings). The battery bank has the autonomy of 163 hours. State of charge (SOC) is almost constant all the time except in the period dry season from June to September. Figure 17 shows the annual monthly average state of charge (SOC) of battery bank.

The technical and financial details for converter are given in Tables 17, 19, and 20. The maximum capacity for the converter is 40 kW. The lifespan for the inverter is assumed to be 25 years. The performance of inverter is assumed to be 90%. The efficiency of rectifier and inverter is 90% and 85%, respectively.

The whole hybrid energy system produces an annual peak power output equivalent to 40 kW except in the period of June and September in which the major source of electricity is solar radiation. Figures 18 and 19 indicate the annual daily profile of the total renewable energy output and

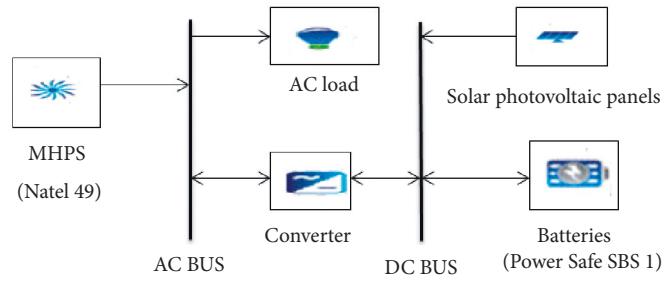


FIGURE 12: Optimal configuration for the proposed off-grid hybrid energy-based system.

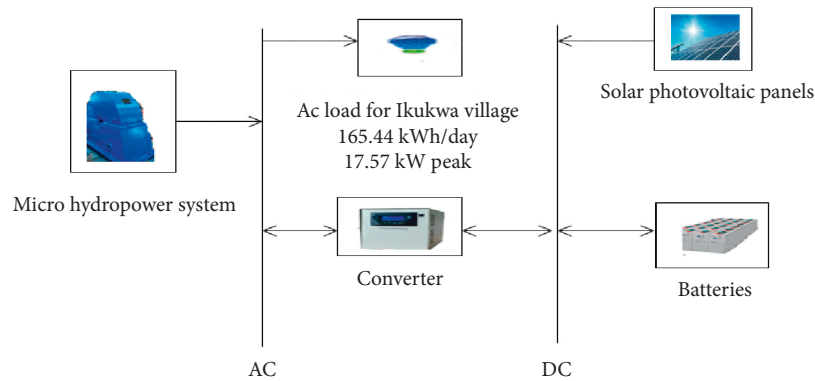


FIGURE 13: Schematic depiction for optimal configuration for the proposed off-grid hybrid energy-based system. Simulation results for technical analysis by HOMER software.

TABLE 17: System architecture.

| Component                              | Capacity in kW/quantity in numbers |
|----------------------------------------|------------------------------------|
| Generic flat-plate photovoltaic panels | 40 kW                              |
| Micro-hydro-turbine/Natel-free jet     | 18.4 kW                            |
| Converter                              | 40 kW                              |
| Batteries/PowerSafe SBS 1500           | 96 nos                             |

TABLE 18: Generation, consumption, and renewable energy utilization.

| Characteristics/factors              | Unit/value | Percentage |
|--------------------------------------|------------|------------|
| <i>Production</i>                    | KWh/year   |            |
| Solar photovoltaic                   | 75 804     | 39.8       |
| Micro-hydro-turbine                  | 114 431    | 60.2       |
| Total                                | 190 235    | 100        |
| <i>Consumption</i>                   | KWh/year   |            |
| AC primary load                      | 60 386     | 100        |
| DC primary load                      | 0          | 0          |
| Deferrable load                      | 0          | 0          |
| Total                                | 60 386     | 100        |
| <i>Quantity</i>                      | KWh/year   |            |
| Excess electricity                   | 128 483    | 67.5       |
| Unmet electrical load                | 0          | 0          |
| Capacity shortage                    | 0          | 0          |
| <i>Quantity of renewable sources</i> | Value      | Unit       |
| Renewable fraction                   | 100        | %          |
| Max. renewable penetration           | 11 347     | %          |

annual scaled data profiles, respectively. During the rainy season, MHPS has a higher generation capacity than SPVS. SPVS with battery energy storage gives power in all seasons

of the year within the proposed hybrid system. In dry season, SPVS is the only alternative energy source of power. Similarly, in the period of dry season battery energy storage

TABLE 19: Economic input parameters for the optimized hybrid renewable energy system.

| Component    | Capacity/quantity (kW/Nos) | Capital cost (\$) | Replacement cost\$ | O&M cost (\$/year) |
|--------------|----------------------------|-------------------|--------------------|--------------------|
| MHT/Natel 49 | 18.4 kW                    | 50 000            | 43000              | 110                |
| Solar panels | 40 kW                      | 2500              | 2500               | 10                 |
| Batteries    | 288 nos                    | 2500              | 2500               | 120                |
| Converter    | 40 kW                      | 1000              | 1000               | 110                |

MHT = micro-hydro-turbine.

TABLE 20: Annualized costs by component.

| Type of cost     | Components of optimized hybrid renewable energy system |               |           |                  | System  |
|------------------|--------------------------------------------------------|---------------|-----------|------------------|---------|
|                  | MHT                                                    | Flat solar PV | Batteries | System converter |         |
| Capital (\$)     | 3867.72                                                | 193.39        | 1, 547    | 77.35            | 5685.55 |
| Replacement (\$) | 0                                                      | 0             | 0         | 32.82            | 32.82   |
| O&M (\$)         | 5000                                                   | 10            | 960       | 110              | 6080    |
| Fuel (\$)        | 0                                                      | 0             | 0         | 0                | 0       |
| Salvage (\$)     | 531.22                                                 | 0             | 281.67    | 6.18             | 819.06  |
| Total (\$)       | 214                                                    | 203.9         | 225.42    | 214              | 10979.4 |

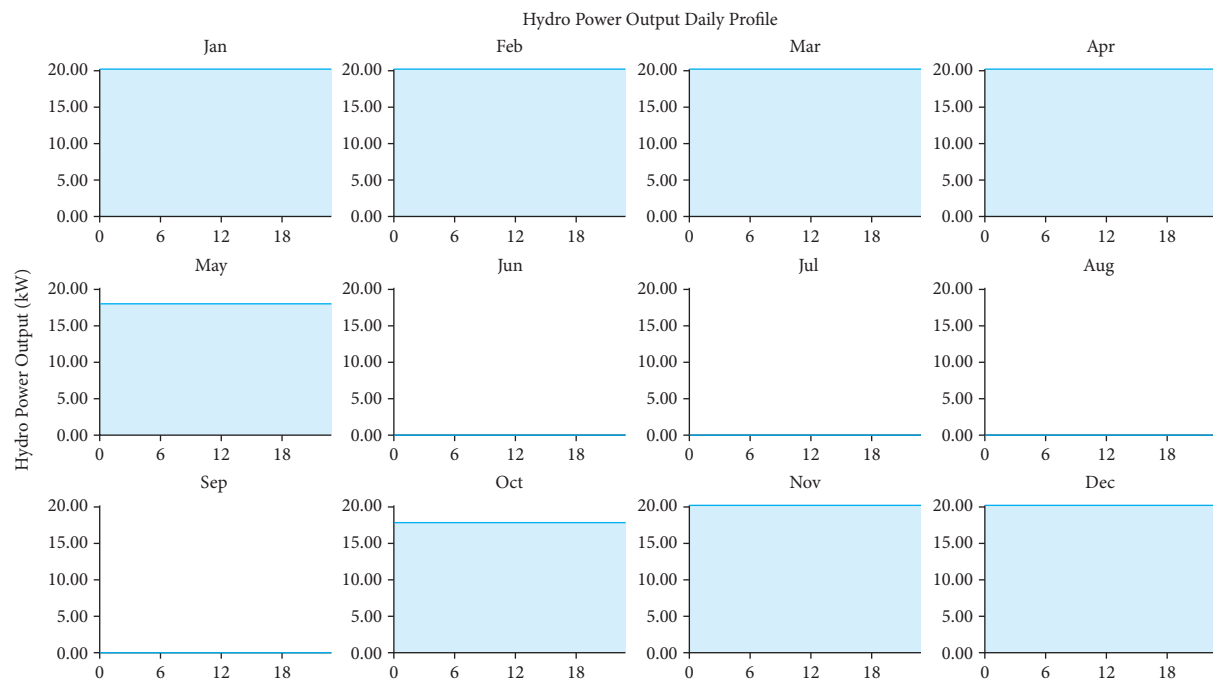


FIGURE 14: Annual daily profile of micro-hydropower output.

demonstrates the highest input power to the system. Figures 20 and 21 illustrate the annual monthly electricity production and hourly power production of the proposed system, respectively. The annualized cost by components of MHT has the highest cost, followed by the cost of batteries, and then, the cost of generic flat solar and the cost of converter are the least. Figure 22 and Table 19 both indicate annualized cost by component in the hybrid system.

This section of this article presents an economic analysis of a 100% renewable energy system consisting of solar photovoltaic panels, micro-hydro-turbine, batteries, and converter. This analysis has been carried out by considering the aforesaid period of the project. The types of annualized and net present costs by components have been analysed.

Various types of costs such as initial cost, replacement cost, operation and maintenance cost, fuel cost, salvage cost, and inflation rate have been considered to be 3.04%. Table 19 shows economic input parameters for the optimized hybrid renewable energy system. The results for the total net present cost, operating cost, and cost of energy are \$ 141, 397.10, \$ 5293.76, and 0.1818/kWh, respectively. Annualized costs by components are indicated in Figure 22 and Table 20, respectively.

**9.1. Limitations of the Proposed Hybrid Energy System.** Apart from the merits of including the flow of the river system, solar PV and batteries in the proposed hybrid system



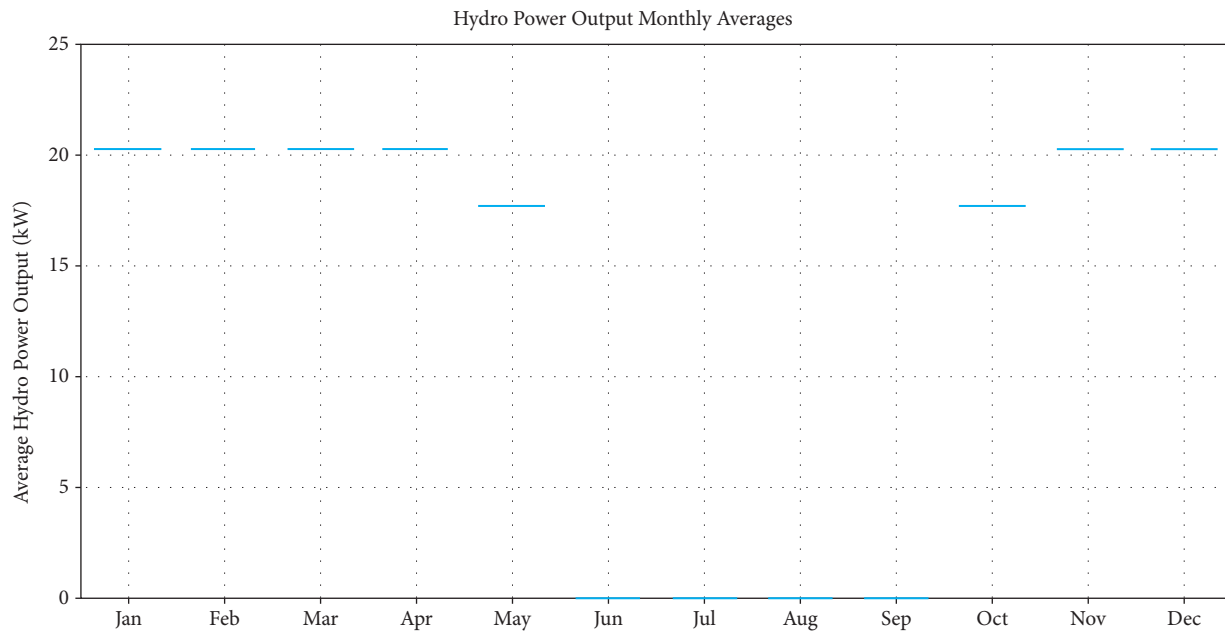


FIGURE 15: Annual monthly averages of hydropower output.

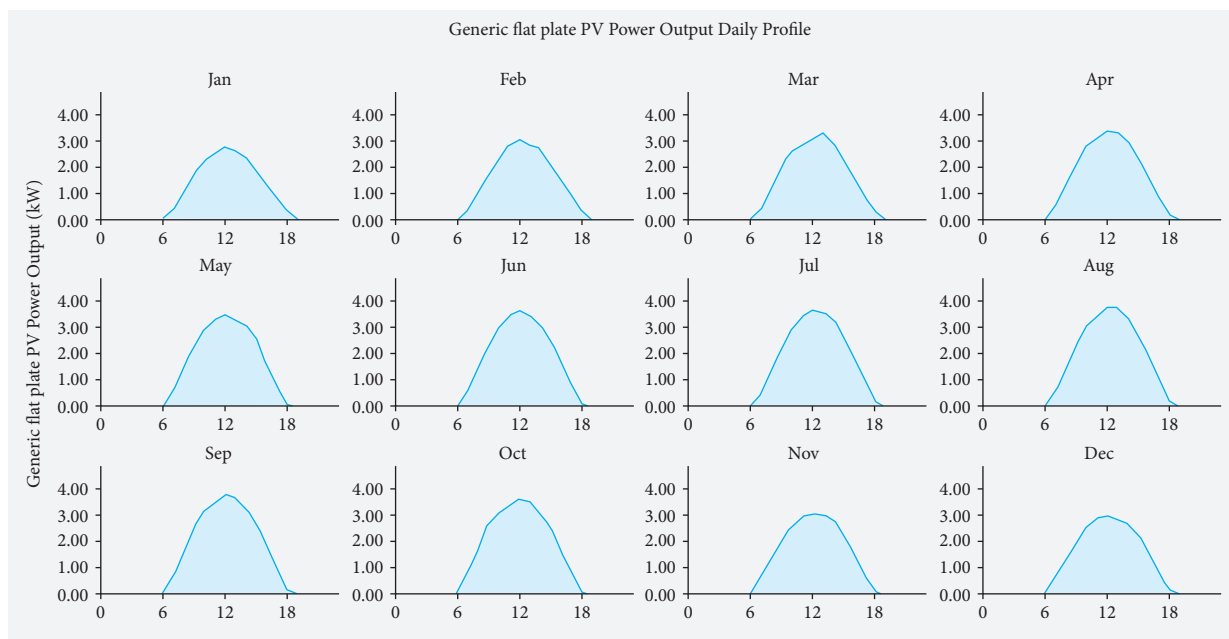


FIGURE 16: Annual daily profile of solar photovoltaic power output. Simulation results for economic analysis by HOMER software.

such as reduced investment cost, running cost (water source and solar energy are free and renewable), time of construction, negative impacts of environment, and power oscillations (inter-day and seasonal variability), the limitations of the hybrid system are described as follows [66]:

- (i) The operation of the hydropower system in nature is a weather-dependent project. Better performance of the flow of the river system depends on the high precipitation in the area of study. The operation of the system may be drastically affected in the

critical dry season. A suitable operation strategy is necessary to avoid abnormalities, particularly during the dry period. The analysis of this specific system does not include control strategy to address the challenge of uncertainty and unpredictability of water flow for the inter-daily, rainy, and dry periods [66].

- (ii) The analysis of the system does not include the dynamics of the solar cells and hydropower excitation system [66].

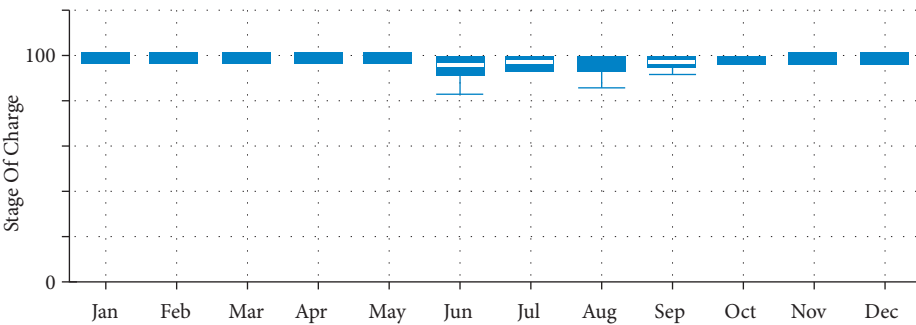


FIGURE 17: Annual monthly average SOC of battery bank.

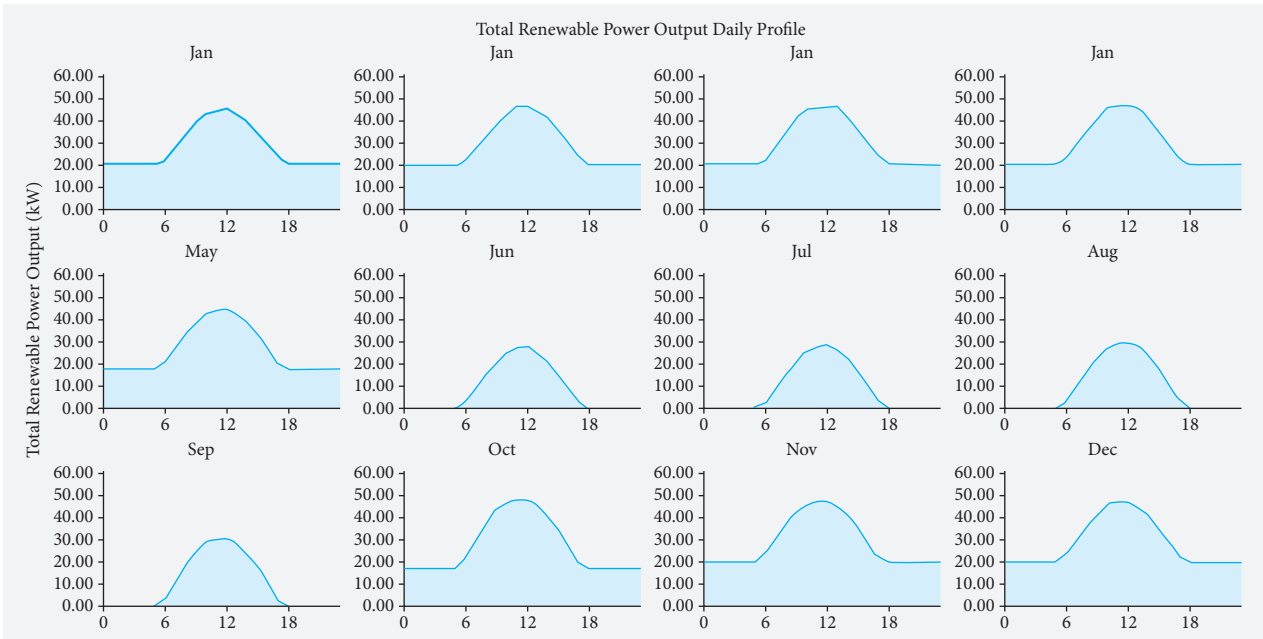


FIGURE 18: Annual daily profile of the total renewable energy.

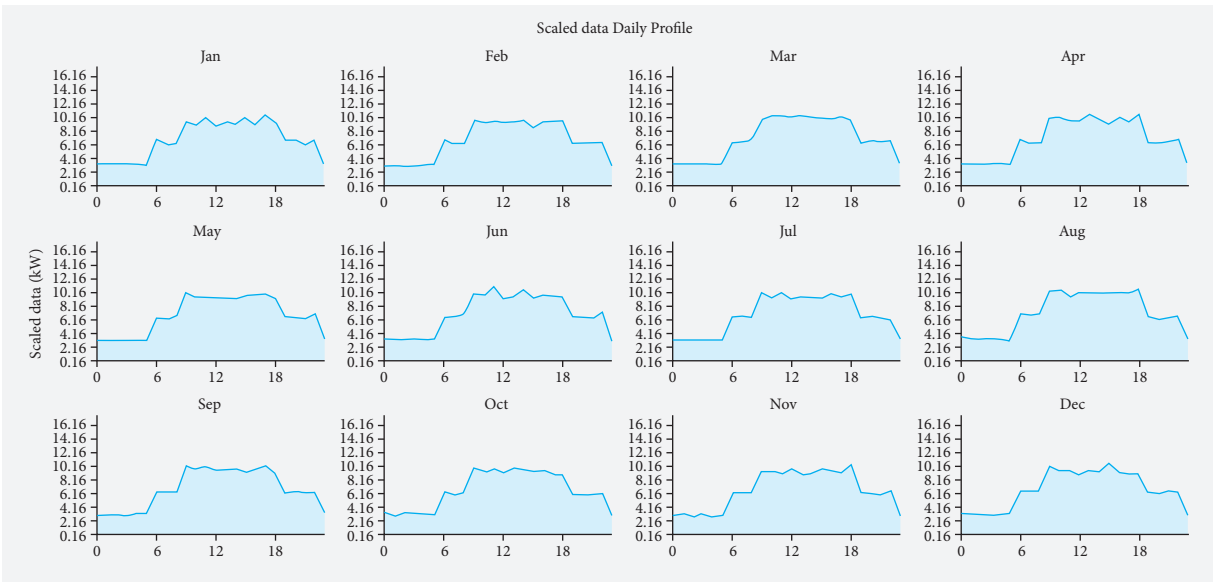


FIGURE 19: Annual scaled data load profile.

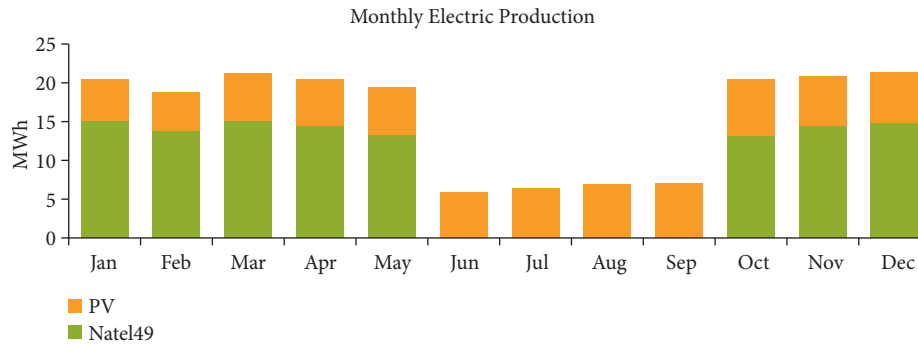


FIGURE 20: Annual monthly electricity production.

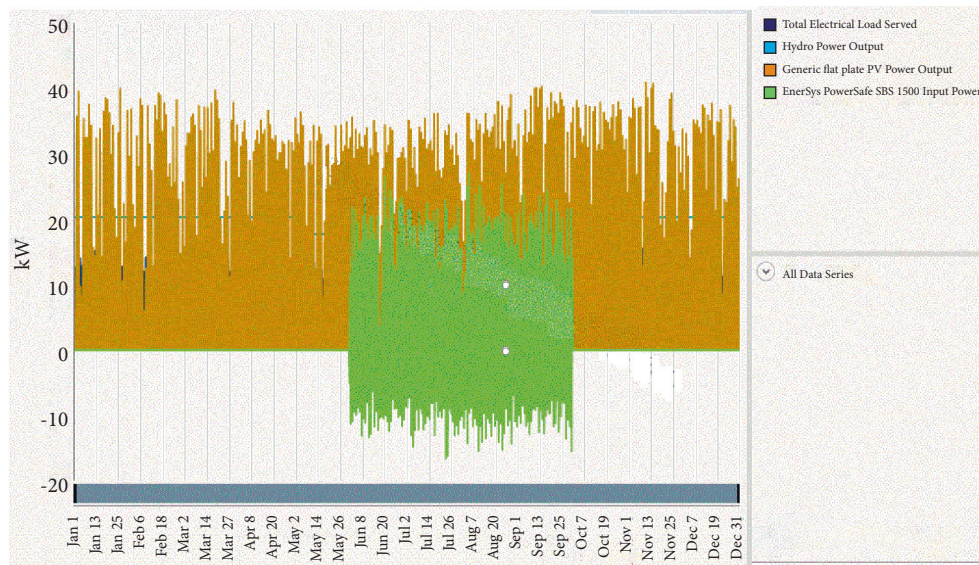


FIGURE 21: Hourly power production of the proposed system.

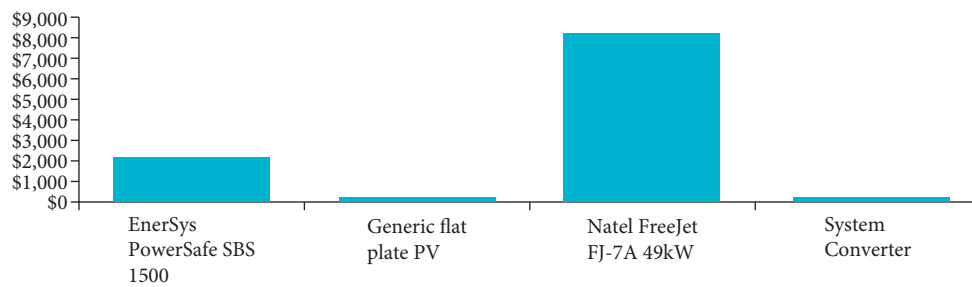


FIGURE 22: Annualized costs by component.

- (iii) The dynamic response of hydro-energy system due to great inertia of water turbine owing to the requirement of the battery energy storage has not been considered in the optimization [56].
- (iv) Fixed efficiency has been used in the optimization process of the system. As water injection from the water turbine affects the efficiency of run of over the river power plant, no control has been included to balance with variable electrical load. This is not

practical as far as water flow is variable due to the fluctuating hydrological situations.

- (v) Application of traditional methods for data measurements may provide inaccurate and unreliable data. In addition, data gathering does not include geological information [66].
- (vi) The analysis of the proposed hybrid system is not grid-connected. The analysis with the addition of grid connection to the hybrid energy system would

be helpful for the minimization of cost incurred by increasing the number of batteries in off-grid decentralized systems [66].

- (vii) Complicated control and limited lifecycle of the battery are subjected to the rainy and dry seasons [66].
- (viii) There is no practical validation of the proposed system to gauge the performance of the system in the real environment due to the economic challenges and limited time.
- (ix) The profile characteristics of the run-of-the river pattern may be affected by weather variations as a result of drastic effect of global warming [66].
- (x) The analysis of the proposed energy system does not involve the environmental component of sustainability. The recommendation that has been described in the conclusion is only based on authors' observation in the area of study [67, 68].

For appropriate performance analysis of the hybrid system, the artificial intelligent-based control strategy should be implemented.

### 9.2. Assessment Criteria of the Proposed Hybrid Energy System.

The proposed hybrid system consisting of MHT, solar PV, and batteries has been optimized using the monthly average flow of water from the tributary under study and global solar radiation. The tributary is located in one of the rural areas of Mbeya Region in Tanzania having prolonged high rainfall. Also, the tributary is well elevated lying in Mount Mbeya having sufficient water flow particularly during the wet season mainly contributed by high rainfall and a few months after the rainy season. The water flow of the tributary is discharged into the River Shongo, which provides water for the livestock and local irrigation schemes of agriculture at Ikukwa Village. The water flow from the tributary is deemed to have a potential of contributing energy to the proposed system in the wet season. The area under study is also exposed to the sufficient coverage of sunlight during the dry season. Solar energy can be harnessed to generate electricity in the hybrid system. However, sunlight is absent at night time and during the rainy season, and therefore, battery energy storage is necessary for covering the excess or shortfall of power supply and for ensuring the fast response of off-grid hydro-energy system. In other words, solar energy source is categorized by uncertainties and huge fluctuations, which can impact adversely the system steadiness and consistency requiring batteries for energy storage. In general, the hydro-energy source provides high power, reliable in wet season, and dries up in the dry season, while solar energy source is intermittent, and thus, these types of energy sources are complementary for ensuring the 100% availability of power supply to meet the load for the whole year. These complementary energy sources are sustainable, renewable, free, and environmentally friendly. The proposed hybrid energy has been analysed using collected average data of available renewable energy sources and electrical demand. Based on the authors' practical experience, literature review and decision-making methods via up-to-date advancement of

using a computer application and AI optimization techniques of the potential of hybrid system have been evaluated, and assessment criteria are summarized as follows [66]:

- (i) The location of study has a high potential of water flowing in the tributary during the wet season and sufficient solar energy source for electricity generation. The energy sources are sustainable and complementary renewable energy sources [67, 68].
- (ii) For the purpose of reliability improvement of the system and cost minimization of batteries, hybridization of renewable energy sources is the best option. Solar and hydro-energy sources are natural-free sources and are also ecologically friendly, and thus, the hybrid power plant combining the ROR power system and solar PV is cost-effective and sustainable. The configuration of the plant has been defined by the lowest NPC [67, 68].
- (iii) Generated power output of the run-of-over river scheme in the optimal hybrid system was estimated using water flow rate, water head, and fixed efficiency. In reality, the dynamics of the hydrological situation always varies over time. Therefore, one of the drawbacks of this method is that it overestimates the actual generated power, particularly run-of-the river systems with low water head due to the limitations of the turbine release, head, and efficiency deviations [67, 68].
- (iv) Power capacities of solar PV and batteries have been estimated according to the monthly profiles of the streamflow, solar radiation, and electrical load profile. The tributary profile was obtained using traditional methods of measurements. Solar radiation data were directly obtained from NASA on the monthly basis, while batteries have been sized according to hourly characteristics for minimization of the hourly fluctuations of solar PV in day time [67, 68].
- (v) Energy requirement was estimated based on rated power capacities of anticipated appliances, and load profile was computed according to the socioeconomic life and norms of the area of study. In the optimization processes of the proposed system, it was assumed that generation capacity greater than the peak maximum accounts for the power losses [67, 68].

**9.3. Optimization Results of the Proposed System Using AI Algorithms.** As it has been mentioned before, the average daily electricity consumption is 511.1 kWh/day (baseline), peak power capacity is 30.31 kW, and load factor is 0.71. For the hybrid electric system to supply sufficient power to satisfy the load, generated electricity should be equal to or greater than the highest power capacity. The proposed system has been optimized when LEPP equals zero denoting no deficiency of power assuming that the hybrid energy system supplies power at 100%. Optimization results that are processed by AI techniques such as PSO, GWO, and GWO-



TABLE 21: Comparative analysis of optimization results of the soft computing and AI methods for the optimized MHPS/solar PV/battery system (LEPP = 0).

| Financial parameters | HOMER      | PSO       | GWO       | GWO-PSOHD |
|----------------------|------------|-----------|-----------|-----------|
| NPC (\$)             | 141 397.76 | 95 167.21 | 92 472.82 | 91,854.10 |
| COE (\$/kWh)         | 0.1818     | 0.1185    | 0.1182    | 0.1181    |

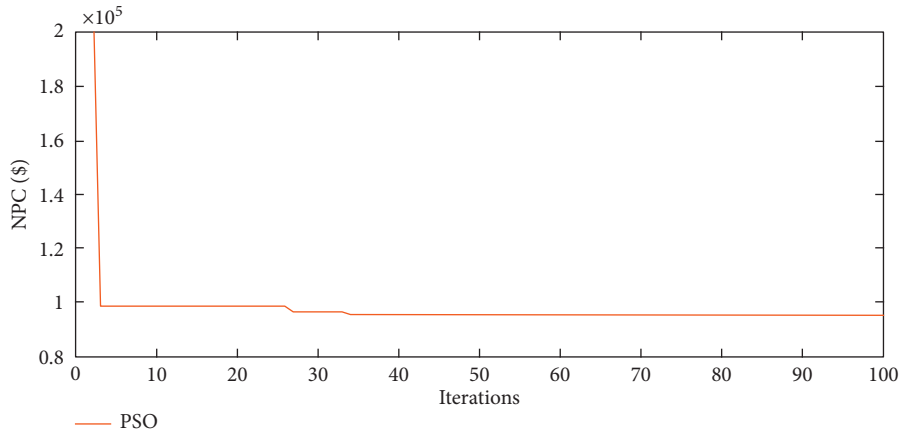


FIGURE 23: NPC optimization for the MHPS/solar PV/battery system using the PSO algorithm.

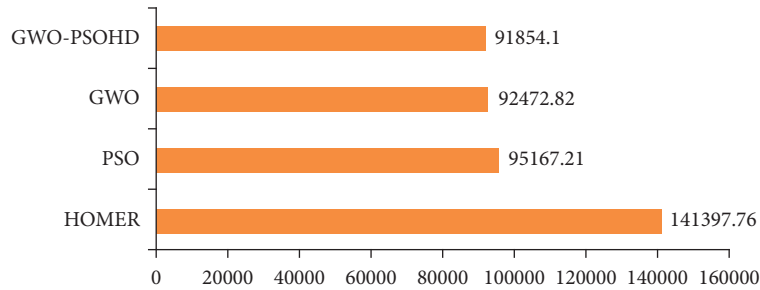


FIGURE 24: NPC for the MHPS/solar PV/battery system by soft computing and AI methods.

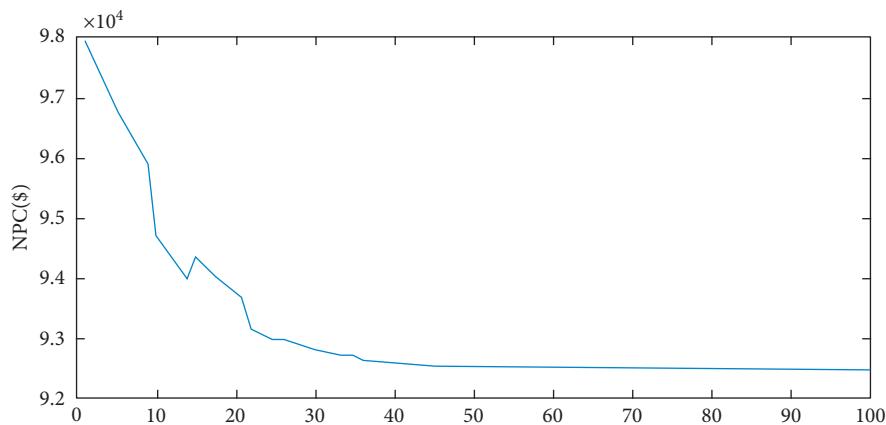


FIGURE 25: NPC optimization for the MHPS/solar PV/battery system using the GWO algorithm.

PSOHD algorithms show the reduction in total NPC and corresponding levelized COE in reference to the soft computing method (HOMER application). Table 21 shows the comparison of the optimization results of the soft

computing and AI methods for the optimized MHPS/solar PV/battery system (LEPP = 0). The AI techniques are implemented in MATLAB, and optimal results are presented as follows.

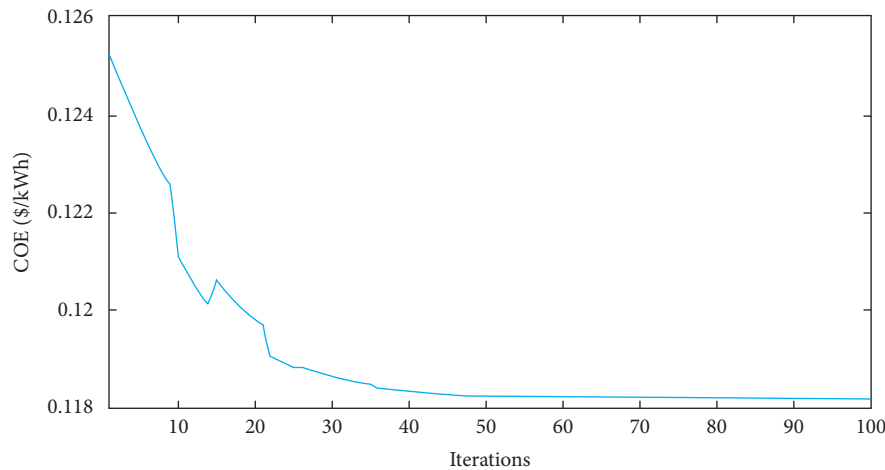


FIGURE 26: COE optimization for the MHPS/solar PV/battery system using the GWO algorithm.

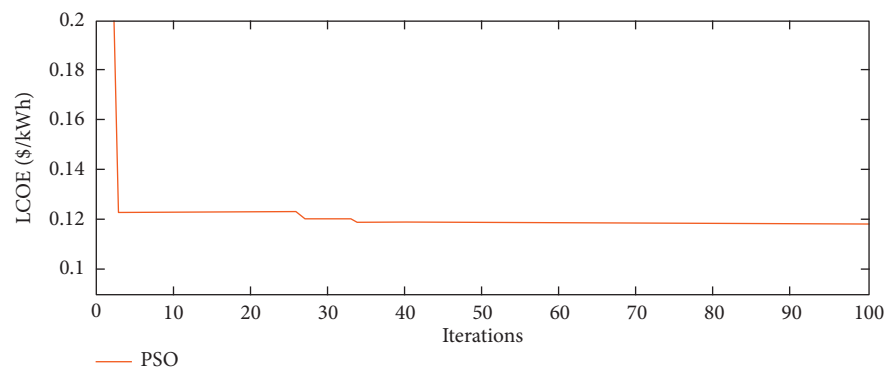


FIGURE 27: COE optimization for the MHPS/solar PV/battery system using the PSO algorithm.

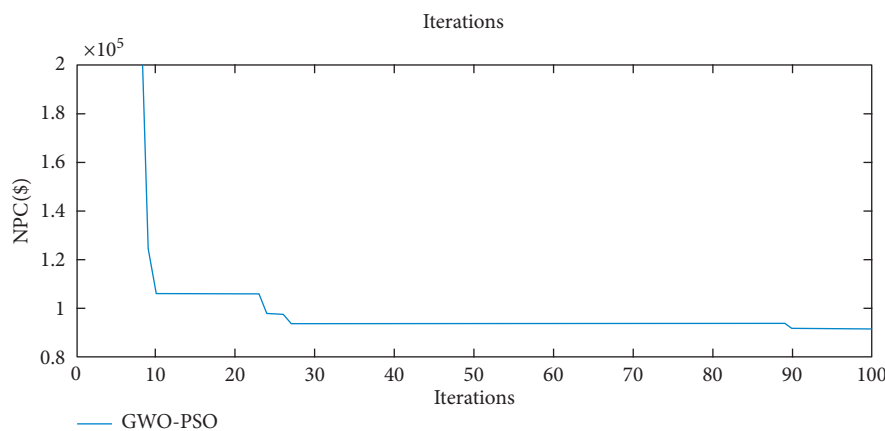


FIGURE 28: NPC optimization for the MHPS/solar PV/battery system using the GWO-PSOHD algorithm.

**9.4. Results of PSO Implementation.** Among the above named three AI optimization techniques, the PSO method has indicated the greatest NPC after HOMER software. The PSO implementation gives accumulated NPC and its corresponding levelized COE, which are \$ 92,472.82 and \$ 0.1182/kWh, respectively. These categories of financial standards are lower in the PSO method than in the HOMER platform.

The NPC and COE are more minimized nearly 67.3%. It denotes that the PSO method offers the superior solution in terms of reduction in cost in comparison with HOMER software although it has the greatest cost in comparison with those costs achieved by the GWO and GWO-PSOHD optimization techniques. Figure 23 describes the NPC optimization results for the MHPS/solar PV/battery system

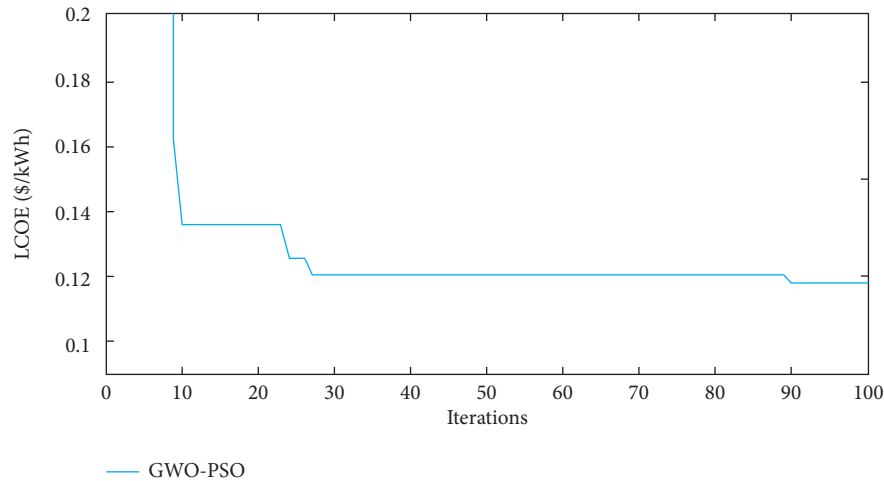


FIGURE 29: COE optimization for the MHPS/solar PV/battery system using the GWO-PSOHD algorithm.

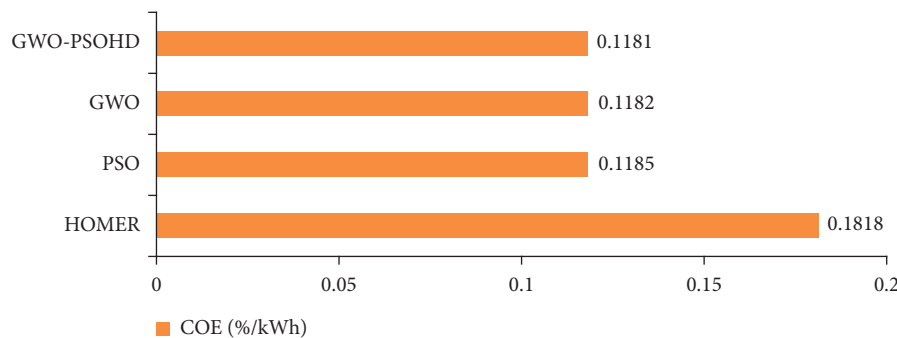


FIGURE 30: COE for the MHPS/solar PV/battery system by soft computing and AI methods.

using the PSO algorithm. Furthermore, Figure 24 illustrates COE optimization for the MHPS/solar PV/battery system using the PSO algorithm.

**9.5. Results of GWO Implementation.** GWO implementation minimizes further accrued NPC and leveled COE of the proposed off-grid hybrid renewable energy-based system having the magnitudes, which are less than that achieved from the PSO method. The optimally designed system using GWO provides NPC and COE, which are \$ 92,472.82 and \$ 0.1182/kWh, respectively. These types of economic values are less in the GWO technique than in the HOMER. Both NPC and COE are further reduced up to around 65%. The GWO techniques offer an intermediate cost between the PSO and GWO-PSOHD techniques having reduced cost in comparison with HOMER software. Figure 25 depicts the NPC optimization results for the MHPS/solar PV/battery system using the GWO algorithm. Furthermore, Figure 26 illustrates COE optimization for the MHPS/solar PV/battery system using the GWO algorithm (Figure 27).

**9.6. Results of GWO-PSOHD Implementation.** The research of this study has employed the GWO-PSOHD algorithm to optimize the proposed energy system. Similarly, this technique has been used to solve the formulated objective

function by the minimization of NPC. The number of iterations and population size is 100 and 50, respectively. Optimal results have indicated that the hybrid algorithm provides the least value of NPC in comparison with other aforementioned optimization methods. Both NPC and COE are more reduced up to about 64.96%. Figure 28 represents the NPC optimization results for the MHPS/solar PV/battery system using the GWO-PSOHD algorithm. Furthermore, Figure 29 illustrates COE optimization for the MHPS/solar PV/battery system using the GWO-PSOHD algorithm.

Moreover, Figures 24 and 30 portray the total NPC and COE for the MHPS/solar PV/battery system of the soft computing and AI optimization methods, respectively. Table 21 illustrates the comparative analysis of the optimization results of the soft computing and AI methods for the optimized MHPS/solar PV/battery system (LEPP = 0).

## 10. Conclusions, Recommendation for the Long-Term Sustainability of the Tributary under Study, and Future Research Work

The techno-economic analysis of unexploited potential of a nonperennial tributary incorporated with solar energy source and batteries has been carried out. A case study has been made

at Ikata tributary located at Ikukwa Village, Mbeya Rural District, Mbeya Region, Tanzania. The main objective is to examine the potential of using the unused tributary hybridized with solar photovoltaic and batteries to meet electrical load demand for small communities without connection to the grid power network. Data for the estimation of water flow rate and head for Ikata tributary have been achieved by direct measurement using traditional methods.

In the micro-hydro/solar photovoltaic/battery system, MHPS immensely indicates a high positive impact in terms of power output. The overall proposed hybrid energy system provides sufficient power, which is mainly contributed by MHPS during the extended period of rainy/winter season. Power capacity for MHPS in the hybrid energy system drops down to zero due to the limitation or absence of water flow in dry period. Within this proposed hybrid energy system, SPVS with batteries provides electricity during rainy and dry seasons. However, SPVS with batteries enormously offers electricity during dry/summer seasons with limited or absence of power output from MHPS.

Three major implications can be made about this research work. First, the hydro-energy source from the tributary has a high abandoned potential of energy, especially in the rural expanses with high and prolonged rainfall. Second, this proposed hybrid energy system is economically feasible with the lowest total NPC and COE. Third, the optimal design of the proposed hybrid system produced by AI optimization techniques (PSO, GWO, GWO-PSOHD) is more cost-effective than that executed by soft computing technique (HOMER Pro software). In the group of AI optimization approaches, the PSO technique has the highest price, while the cost acquired by the GWO-PSOHD technique is the lowest, and the cost acquired by the GWO method lies between the two.

In the survey made by the authors at the site of the project, much deforestation was observed. Villagers get firewood for cooking food and heating purposes by cutting trees in Mbeya Mountains. The ongoing deforestation should be stopped for achieving the long-term sustainability of Ikata tributary and other hydro-energy sources [23, 69]. The action of stopping the deforestation can also sustain the performance of MHPS in future.

In this study, the traditional approach in direct measurements of power has been applied and used for the mathematical analysis of theoretical power of MHPS. Some of the parameters obtained by such approach, for instance, water flow rate, have been used as a reference during the simulation process by HOMER Pro software. Therefore, in future the research can be furthered by improved methodological approaches including more modern AI methods while integrating other energy sources with the system under investigation. The detailed techno-economic feasibility analysis can be implemented using advanced methods of measuring techniques, geological survey, and grid integration to support the national power network from TANESCO. Also, this study can be expanded by implementing sensitivity analysis under the varied LEPP on the assumption that the hybrid renewable energy-based system does not produce power at 100% [70].

## Data Availability

Data are available on the request from the corresponding author.

## Conflicts of Interest

The authors declare that there are no conflicts of interest.

## Authors' Contributions

IJM collected the data in Tanzania, performed the analysis, and wrote the manuscript. MR and NK provided valuable inputs for better output of research work. All authors read and approved the manuscript.

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